

Low-rank random-feature networks: depth–rank scaling and conditioning in the NTK regime

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Abstract

We study low-rank random-feature networks (RF-LR), where each layer is a frozen random feature map followed by a trainable readout into a bottleneck of dimension $r \ll N$. We focus on a finite-sample kernel regime: we take a sequential infinite-width limit $N \rightarrow \infty$ at fixed dataset size n , and then study how the remaining bottleneck randomness concentrates as $r \rightarrow \infty$.

Our main results quantify how depth L and rank r control conditioning and optimization scales in kernel regression. In an exactly solvable equicorrelated dataset model we obtain an explicit Gram spectrum, with a nontrivial scale $\Theta(1/(rL))$ on $\mathbf{1}^\perp$, yielding a kernel time scale $\Theta(P/n)$ under $P \asymp LrN$ and $N \asymp n^2$. We also derive an explicit RF-LR NTK recursion and show, as a secondary consequence, that the mean three-layer kernel induces the same RKHS as the shallow ReLU kernel via an endpoint (Puiseux) expansion. We do not analyze the empirical RF-LR NTK in the quadratic random-matrix scaling $n \rightarrow \infty$; adapting resolvent-based bulk-law techniques developed for shallow two-layer NTKs Benigni and Paquette (2025) to low-rank RF-LR and deeper architectures is left for future work.

Keywords: Neural Tangent Kernel, Random Features, Low Rank, Depth Scaling, Conditioning

1. Introduction

Low-rank random-feature networks (RF-LR) are architectures where each layer is implemented as a random feature map followed by a trainable linear readout into a low-dimensional bottleneck Zhang et al. (2025). Concretely, for each layer we sample frozen random feature directions $w^{(\ell)}$ and train the readouts $A^{(\ell)}$ (and biases), which map the feature vector into a bottleneck of dimension r . This reduces the trainable parameter count from $O(N^2)$ (dense layers) to $O(rN)$ while keeping the random-feature geometry explicit.

Low-rank parameterizations and factorizations have long been used to reduce memory and computation in neural networks, and to tailor optimization under explicit parameter budgets Sainath et al. (2013); Novikov et al. (2015); Arora et al. (2018); Kamalakara et al. (2022). In such models one can often separate an upstream representation map from a downstream linear predictor, and the choice of which component is learned can materially affect both optimization and generalization. Recent information-theoretic generalization bounds based on conditional mutual information suggest that stochastic projection/quantization combined with learning a downstream predictor can yield tighter guarantees, providing further motivation for architectures that keep the “left” projection map fixed (or randomized) while training the “right” readout map Sefidgaran et al. (2025). Recent empirical work by Zhang et al. Zhang et al. (2025) has demonstrated that RF-LRs achieve global strong-convergence across a wide range of tasks, including neural-network based PDE solving with and without Physics-Informed losses Raissi et al. (2017). Empirically, training can display stepwise dynamics (plateaus separated by sharp drops) in some regimes Zhang et al. (2025). This paper does not

aim to explain these phenomena beyond the kernel regime; instead we develop an NTK-level theory for RF-LR at initialization and in the lazy/NTK regime, focusing on conditioning/optimization scales and on concentration in the bottleneck dimension r .

The Neural Tangent Kernel (NTK) framework [Jacot et al. \(2018\)](#) provides the natural lens for analyzing the early training stage. This perspective has been extended to finite width [Huang and Yau \(2019\)](#); [Dyer and Gur-Ari \(2019\)](#) and large-depth [Guillen et al. \(2025\)](#); [Yang et al. \(2023\)](#); [Terjék \(2022\)](#), revealing how the Laplacian kernel structure governs the gradient descent (GD) training dynamics and generalization. Applying NTK theory to RF-LR raises specific technical questions: the kernel recursion differs from the standard MLP recursion because only the readouts (and biases) are trained, and the bottleneck dimension r induces additional concentration and scaling effects.

Main results (informal).

Theorem 1 (Main results in the NTK regime (informal)) *Consider RF-LR with ReLU features under sequential infinite width $N \rightarrow \infty$ and bottleneck dimension r . In the lazy/NTK regime, gradient flow reduces to kernel regression with an (approximately) fixed kernel [Jacot et al. \(2018\)](#). Our main results are:*

- **Recursion.** *The RF-LR NTK satisfies an explicit infinite-width recursion with an explicit $1/r$ bottleneck factor, yielding a closed-form expansion ([Theorem 4](#) and [Corollary 5](#)).*
- **Main: depth–rank scaling and optimization scale.** *In an equicorrelated dataset model, the Gram spectrum is explicit and the nontrivial spectrum on $\mathbf{1}^\perp$ has scale $\lambda_\perp = \Theta(1/(rL))$ ([Proposition 9](#)); for centered targets and $1/n$ -normalized square loss this yields an optimization time scale $\Theta(n/\lambda_\perp) = \Theta(nrL)$, which becomes $\Theta(P/n)$ under the budget scaling $P \asymp LrN$ and $N \asymp n^2$ ([Corollary 11](#)).*
- **RKHS (three layers; secondary consequence).** *The mean three-layer kernel induces the same RKHS as the shallow ReLU kernel ([Corollary 16](#)).*

We adopt the standard NTK setup (ReLU activation, isotropic Gaussian initialization, and a sequential infinite-width limit). Accordingly, we analyze the induced kernel at initialization and its deterministic limits.

1.1. Architecture

The considered architecture is a stacking of high then low width layers:

The loss landscape of neural networks is closely related to generalization and trainability, and in the kernel regime the spectral properties of the Gram matrix control optimization rates and spectral bias [Jacot et al. \(2018\)](#); [Cao et al. \(2020\)](#). We focus on this kernel-regime viewpoint for RF-LR by characterizing the limiting kernels and their spectra at initialization under explicit assumptions.

Our analysis views intermediate-layer random variables as outputs of a composition of (conditional) Gaussian processes in the sequential infinite-width limit, which enables explicit computations of mean kernels and fluctuations.

Guiding question.

Is low rank all we need, and how does optimization in neural networks interact with the curse of dimensionality?

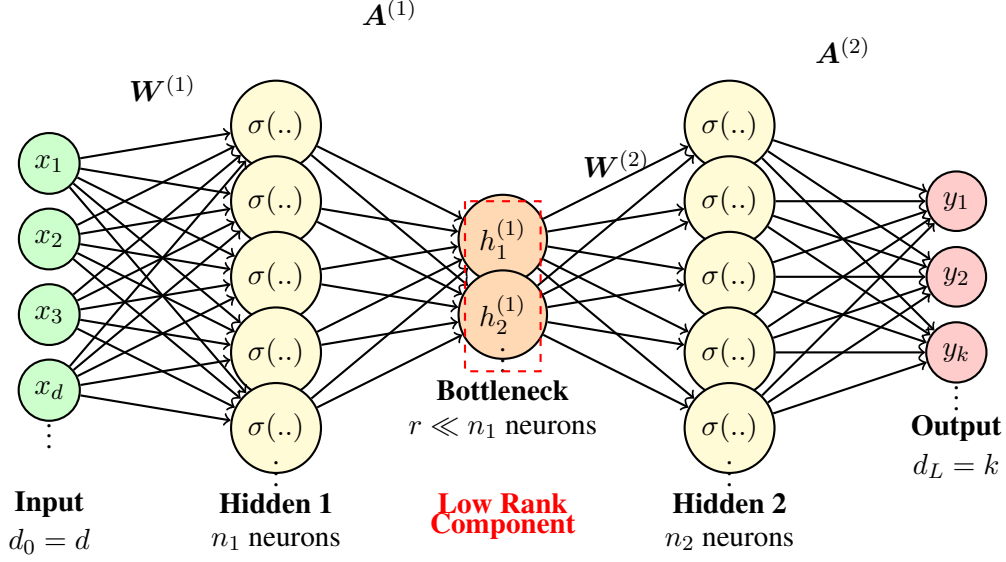


Figure 1: Architecture of a low-rank random-feature network (RF-LR). The red dashed connections indicate the low-rank component that reduces dimensionality from n_1 to $r \ll n_1$ neurons. We train the readouts (and biases) while keeping the feature directions frozen.

This question is closely related to recent discussions of high-dimensional obstacles in neural network optimization [Na and Yang \(2025\)](#). In the NTK regime, the relevant function class is the RKHS induced by the limiting kernel; in particular, low-rank structure does not remove the high-dimensional spectral decay inherent to ReLU kernels on the sphere (Corollary 16).

Progress on the open question. Recent work has begun to clarify which aspects of optimization degrade with dimension and which admit dimension-robust descriptions. On the negative side, explicit high-dimensional obstruction results indicate that optimization can become increasingly constrained as the ambient dimension grows, even under benign data models [Na and Yang \(2025\)](#). On the positive side, particle and optimal-transport viewpoints provide regimes where the loss landscape and dynamics admit tractable characterizations, including asymptotic convexity phenomena and dimension-free bounds in two-layer settings [Rotskoff and Vanden-Eijnden \(2018\)](#); [Mei et al. \(2018, 2019\)](#); [Chizat and Bach \(2018\)](#). In the kernel regime, recent quantitative bounds on the smallest NTK eigenvalue for deep ReLU networks connect data/width scaling to optimization rates [Nguyen et al. \(2022\)](#). Complementary perspectives based on dynamical invariants and “central flow” structure further suggest organizing principles for optimization in deep models beyond the strictly kernel setting [Cohen et al. \(2025\)](#).

Optimization message and technical approach. The conditioning/optimization scaling law and the RKHS identification are summarized in Theorem 1; the technical ingredients are outlined in Appendix A.6.

1.2. Related work

The Neural Tangent Kernel (NTK) framework [Jacot et al. \(2018\)](#) characterizes infinitely wide neural networks trained by gradient descent as kernel methods with a fixed kernel. Extensions cover finite-width corrections [Arora et al. \(2019\)](#); [Dyer and Gur-Ari \(2019\)](#), depth scaling [Yang \(2019\)](#); [Terjék \(2022\)](#), and random-matrix limits for Gram spectra in quadratic regimes [Benigni and Paqueotte \(2025\)](#).

Low-rank and factorized architectures reduce trainable parameters and can change optimization dynamics [Sainath et al. \(2013\)](#); [Novikov et al. \(2015\)](#); [Arora et al. \(2018\)](#). In RF-LR/MMNN, freezing the feature directions and training only readouts (and biases) isolates the bottleneck dimension r as a natural control parameter for kernel concentration and depth scaling.

From the standpoint of generalization, stochastic projections and compression mechanisms can be beneficial when combined with learning a final predictor; see, e.g., the conditional mutual information analysis of [Sefidgaran et al. \(2025\)](#). This supports studying regimes where upstream random feature maps are kept fixed and the downstream readout is trained.

Low-rank adaptation (LoRA) is a widely used parameter-efficient fine-tuning method for foundation models, where training is restricted to a low-rank update of a pre-trained weight matrix. Recent theory begins to locate LoRA between the kernel/lazy regime and full feature learning: in a student–teacher setting, [Dayi and Chen \(2024\)](#) shows that low-rank fine-tuning can exhibit nonlinear dynamics beyond linearized kernels while remaining efficiently learnable under online gradient descent. Complementarily, in the NTK regime, [Jang et al. \(2024\)](#) establishes that sufficiently large LoRA rank can eliminate spurious local minima and enables gradient descent to find low-rank solutions. These results reinforce the broader motivation for understanding which parts of a factorized parameterization should be trained and how low-rank constraints shape optimization and generalization.

For rotation-invariant ReLU kernels, the “deep equals shallow” phenomenon and RKHS characterization via endpoint expansions were established in [Bietti and Bach \(2021\)](#). We leverage this framework in the RF-LR setting by combining Fisher–Kibble decoupling [Fisher \(1915\)](#); [Hotelling \(1953\)](#) with a hypergeometric/Puiseux analysis of the mean kernel near $\rho = 1$.

1.3. Our contributions

Theorem 1 summarizes our main contributions; the formal statements are Theorem 4, Theorem 7, Proposition 9, Corollary 11, and Corollary 16. Our main techniques and proof sketch are summarized in Appendix A.6.

Scope and guarantees (what we analyze and what we do not claim).

- **Objects analyzed.** We study (i) the RF-LR NTK recursion in the sequential infinite-width limit (Theorem 4); (ii) an exact depth–rank scaling theorem for the Gram spectrum in an equicorrelated dataset model (Section 4); and (iii) rank-driven concentration and mean-kernel consequences in the three-layer case (Section 5 and Appendix A.6).
- **Optimization in-scope.** In the lazy/NTK regime, gradient flow reduces to kernel regression with an (approximately) fixed kernel (Theorem 4). Convergence rates are governed by the training-set Gram spectrum; our contribution is to give explicit depth–rank scaling predictions for this spectrum in RF-LR.

- **Not claimed.** We do not analyze feature learning (time evolution of the NTK), adaptive optimizers, or continual-learning/forgetting phenomena. Any references to such effects are purely motivational and are not used as claims or evidence.

Regimes. We work in a finite-sample kernel regime (*Regime 1*): we fix a dataset of size n , take a sequential infinite-width limit $N \rightarrow \infty$, and then study how the remaining bottleneck randomness concentrates as $r \rightarrow \infty$. We *do not* pursue the quadratic random-matrix regime $n \rightarrow \infty$ and bulk spectral laws for the *empirical* RF-LR NTK here (*Regime 2*). Bulk laws in this scaling have recently been established for (shallow) two-layer NTKs via resolvent replacement and Gaussian-equivalence methods [Benigni and Paquette \(2025\)](#); extending such techniques to the RF-LR low-rank setting and to depth $L \geq 2$ is left for future work.

2. Network definition and EOC parameterization

2.1. RF-LR architecture

Let $\mathbf{h}^{(0)}(x) = x \in \mathbb{R}^{d_0}$. For $\ell = 1, \dots, L$,

$$\mathbf{h}^{(\ell)}(x) = \frac{1}{\sqrt{n_\ell}} \sum_{j=1}^{n_\ell} A_j^{(\ell)} \sigma\left(w_j^{(\ell)\top} \mathbf{h}^{(\ell-1)}(x)\right) + c^{(\ell)}. \quad (1)$$

Training policy: The readouts $A^{(\ell)}$ and layer biases $c^{(\ell)} \in \mathbb{R}^{d_\ell}$ are trained; the feature directions $w^{(\ell)}$ are frozen random draws (i.i.d. Gaussian). The $1/\sqrt{n_\ell}$ scaling (NTK parameterization) ensures a well-defined infinite-width limit.

Width vs bottleneck dimension. We distinguish the feature width n_ℓ (number of random features) from the output dimension $d_\ell = \dim(\mathbf{h}^{(\ell)})$. The trainable matrix $A^{(\ell)} \in \mathbb{R}^{d_\ell \times n_\ell}$ has columns $A_j^{(\ell)} \in \mathbb{R}^{d_\ell}$. A *bottleneck* layer corresponds to $d_\ell = r \ll n_\ell$, yielding $O(r n_\ell)$ trainable parameters and $\text{rank}(A^{(\ell)}) \leq r$ automatically.

Training parameters, biases, and initialization. Initialization is

$$w_j^{(\ell)} \sim \mathcal{N}(0, I_{d_{\ell-1}}/d_{\ell-1}), \quad A_{ij}^{(\ell)} \sim \mathcal{N}(0, \sigma_A^2), \quad c^{(\ell)} = 0,$$

We use the *edge-of-chaos (EOC)* scaling for the weights so that $\text{Cov}(w_j^{(\ell)}) = I_{d_{\ell-1}}/d_{\ell-1}$ [Hayou et al. \(2018\)](#). In our bottleneck regime we take $d_{\ell-1} = r$ for $\ell \geq 2$, so $w_j^{(\ell)} \in \mathbb{R}^r$ with covariance I_r/r .

We include a layer bias $c^{(\ell)}$ for completeness, but we avoid introducing extra scaling parameters for it. Bias scaling can be absorbed into the learning rate; the infinite-width and NTK structures remain qualitatively identical, and the kernel-regime predictions depend on kernel geometry, not absolute scales.

With this setup in place, we record the standing assumptions and conventions used throughout the paper.

2.2. Assumptions

We collect the standing assumptions and conventions used throughout the paper; expanded discussion is in Appendix A.

- **Sequential infinite width.** We take $n_1 \rightarrow \infty, \dots, n_L \rightarrow \infty$ with i.i.d. Gaussian random features, frozen after initialization, under NTK scaling $1/\sqrt{n_\ell}$ Jacot et al. (2018) (Appendix A.1).
- **Bottleneck regime and scaling.** For $\ell \geq 2$, $d_\ell = r \ll n_\ell$, and we use edge-of-chaos scaling $\text{Cov}(w_j^{(\ell)}) = I_r/r$ (Appendix A.2).
- **Input normalization.** For RKHS statements we focus on normalized inputs (typically $x \in \mathbb{S}^{d-1}$), so kernels are zonal functions of the cosine similarity ρ ; concentration statements keep explicit $\|x\|\|x'\|$ factors when needed.

Homogeneous activation. We work under the positive 1-homogeneity assumption (satisfied by σ); see Assumption 1 in Appendix A. This condition is used for Fisher–Kibble decoupling (Lemma 12) and for separating radial and angular fluctuations in the three-layer kernel.

2.3. Parameterization, signal propagation, and kernels

We briefly record the scaling conventions used in the kernel calculations; details are in Appendix A.

EOC scaling. We choose the variance σ_A so that the forward variance remains stable across depth (edge of chaos); the resulting recursion for the layerwise variance $q^{(\ell)}$ and the condition on σ_A are given in Appendix A.2.

Bottleneck scaling and the $1/r$ factor. In the bottleneck regime $d_{\ell-1} = r$ for $\ell \geq 2$, the EOC scaling uses $\text{Cov}(w) = I_r/r$, which keeps base and derivative kernels $O(1)$. The RF-LR NTK recursion carries an explicit prefactor $1/r$ at bottleneck layers; see Appendix A.2 and Appendix B.1.

With an abuse of notation, we denote by $\Sigma^{(\ell)}$ the base kernel and by $\dot{\Sigma}^{(\ell)}$ the derivative kernel.

Definition 2 (Base and derivative kernels) *In the sequential infinite-width limit, the layer- ℓ base kernel is*

$$\Sigma^{(\ell)}(x, x') = \mathbb{E}_w \left[\sigma(w^\top \mathbf{h}^{(\ell-1)}(x)) \sigma(w^\top \mathbf{h}^{(\ell-1)}(x')) \right], \quad (2)$$

and the derivative kernel is

$$\dot{\Sigma}^{(\ell)}(x, x') = \mathbb{E}_w \left[\dot{\sigma}(w^\top \mathbf{h}^{(\ell-1)}(x)) \dot{\sigma}(w^\top \mathbf{h}^{(\ell-1)}(x')) \|w\|^2 \right]. \quad (3)$$

For $\ell = 1$ and centered Gaussian w , $\Sigma^{(1)}$ is rotation-invariant with standard closed forms in terms of input cosine similarity $\rho = \langle x, x' \rangle / (\|x\|\|x'\|)$:

$$\Sigma^{(1)}(x, x') = \frac{\|x\|\|x'\|}{2\pi} ((\pi - \theta) \cos \theta + \sin \theta), \quad \theta = \arccos(\rho).$$

We emphasize that these are not the NNGP kernel and the classical derivative kernel appearing in the NTK formulation. Rather, they are the base kernel and the derivative kernel of the Gaussian process $\mathbf{h}^{(\ell)}$ conditional on $\mathbf{h}^{(\ell-1)}$. By rotation invariance of Gaussian weights, these kernels depend on inputs only through the cosine similarity $\rho = \langle x, x' \rangle / (\|x\| \|x'\|)$. We will therefore write $\Sigma^{(\ell)}(\rho)$ and $\dot{\Sigma}^{(\ell)}(\rho)$. When no confusion arises (e.g., for $\ell = 1$), we will also abuse notation and write $\sigma(\rho)$ for the scalar kernel function; this σ denotes the kernel-as-a-function-of- ρ , not the activation σ . We can now state the base and derivative kernels precisely.

Given these definitions, we can now state the following composition result.

Theorem 3 (Gaussian process composition) *In the sequential infinite-width limit, the hidden states $\mathbf{h}^{(\ell)}$ form a composition of Gaussian processes:*

- *Conditionally on $\mathbf{h}^{(\ell-1)}$, for any finite collection of inputs $\{x_1, \dots, x_m\}$, the vector $(\mathbf{h}^{(\ell)}(x_1), \dots, \mathbf{h}^{(\ell)}(x_m))$ converges in distribution to a multivariate Gaussian.*
- *The conditional limiting Gaussian has mean zero and covariance matrix $[\Sigma^{(\ell)}(x_i, x_j)]_{i,j=1}^m$, where $\Sigma^{(\ell)}$ is the base kernel defined in Definition 2 and depends on $\mathbf{h}^{(\ell-1)}$*

Proof See Appendix B.2. ■

3. NTK recursion

We now state the main result characterizing the NTK recursion for MMNNs. This recursion governs how the kernel evolves through layers and determines the training dynamics in the infinite-width limit.

Theorem 4 (Infinite-width NTK recursion) *Under NTK parameterization $1/\sqrt{n_\ell}$ and a sequential limit $n_1 \rightarrow \infty, \dots, n_L \rightarrow \infty$ with $\mathbf{w}^{(\ell)} \sim \mathcal{N}(0, \mathbf{I}_{d_{\ell-1}/d_{\ell-1}})$ i.i.d. frozen after initialization, the layer- ℓ NTK satisfies*

$$\Theta^{(0)}(x, x') = 0, \tag{4}$$

$$\Theta^{(1)}(x, x') = 1 + \Sigma^{(1)}(x, x'), \tag{5}$$

$$\Theta^{(\ell)}(x, x') = 1 + \frac{1}{r} \Theta^{(\ell-1)}(x, x') \cdot \dot{\Sigma}^{(\ell)}(x, x') + \frac{1}{r} \Sigma^{(\ell)}(x, x'), \quad \ell \geq 2, \tag{6}$$

where the factor $1/r$ appears for bottleneck layers $\ell \geq 2$ (low-rank regime). For $\ell = 1$, $\Sigma^{(1)}$ is deterministic; for $\ell > 1$, $\Sigma^{(\ell)}$ and $\dot{\Sigma}^{(\ell)}$ are random fields whose fluctuations shrink with the rank r of bottlenecks. The additive constant 1 term comes from the trained layer biases $c^{(\ell)}$ and corresponds to the constant mode (which vanishes after centering).

Furthermore, the gradient flow dynamics in the NTK regime satisfy

$$\frac{d}{dt} f_t(x) = - \int \Theta^{(L)}(x, x') (f_t(x') - y(x')) d\mu(x'),$$

where μ is the data distribution and y is the target function.

Proof: See Appendix B.1, Proof B.1.

We can now derive an explicit closed-form expression for the NTK at any depth by expanding the recursion.

Corollary 5 (General L-layer NTK; closed form) *Under the assumptions of Theorem 4, for any $L \geq 1$, the NTK admits the explicit form:*

$$\Theta^{(L)}(x, x') = \sum_{\ell=1}^L \frac{1}{r^{L-\ell}} \left(\prod_{k=\ell+1}^L \dot{\Sigma}^{(k)}(x, x') \right) + \sum_{\ell=1}^L \frac{1}{r^{\max(1, L-\ell)}} \Sigma^{(\ell)}(x, x') \prod_{k=\ell+1}^L \dot{\Sigma}^{(k)}(x, x'), \quad (7)$$

where the first sum has bias contributions (empty product = 1 when $\ell = L$) and the second has fresh-basis contributions; each bottleneck layer contributes a factor $1/r$. For $L = 2$: $\Theta^{(2)} = 1 + \frac{1}{r}(\dot{\Sigma}^{(2)} + \Sigma^{(1)}\dot{\Sigma}^{(2)} + \Sigma^{(2)})$.

Proof Expanding the recursion in Theorem 4 yields the two sums in (7). A complete proof by induction (including an explicit $L = 3$ expansion) is given in Appendix B.4; see also Appendix 18. ■

Finite-width effects. Beyond the infinite-width limit, one can in principle study finite-width corrections to the kernel via cumulant/resolvent expansions or diagrammatic methods; see Dyer and Gur-Ari (2019) for a Feynman-diagram approach in wide-network asymptotics. We do not pursue these corrections here.

Relation to fully trained MLPs at the edge of chaos. For comparison with the fully trained (full-width) MLP NTK at EOC, see Appendix A.3 and the reference closed form in Proposition 19.

3.1. Large-depth regime: probabilistic kernel and exponential decay

The mean (probabilistic) recursion and the resulting nested conditional expectations are essential for understanding the deep ($L \rightarrow \infty$) behavior, but the full discussion is lengthy. We therefore defer the probabilistic recursion, its nested-expectation form, and the exponential depth-suppression bound to Appendix A.4 (see also Appendix C.3 for related open problems).

We now specialize to the three-layer case, where we can obtain a particularly compact form under a homogeneity assumption on the activation function.

Corollary 6 (three-layer NTK: compact form) *Under the assumptions of Theorem 4 with $L = 2$,*

$$\begin{aligned} \Theta^{(1)}(x, x') &= 1 + \Sigma^{(1)}(x, x'), \\ \Theta^{(2)}(x, x') &= 1 + \frac{1}{r} \Theta^{(1)}(x, x') \cdot \dot{\Sigma}^{(2)}(x, x') + \frac{1}{r} \Sigma^{(2)}(x, x'). \end{aligned}$$

Proof Immediate from Theorem 4. ■

4. Depth scaling of the RF-LR NTK

4.1. Main results

We now state our main depth–rank scaling results for the mean RF-LR NTK and their dataset-level consequences. The recursion contains an explicit $1/r$ bottleneck factor, which leads to saturation of the kernel magnitude as depth grows, while the diagonal–off-diagonal gap decays with depth in the mean recursion; a principal consequence is an explicit spectrum formula in an equicorrelated Gram model (Proposition 9).

Preliminaries (appendix). The scalar ReLU kernel expansions and the deterministic proxy recursion used below are collected in Appendix D.4.

Theorem 7 (Depth scaling for the mean RF-LR NTK) *Fix $r > 1$ and $\rho_1 \in (-1, 1)$. Let $\Theta^{(k)}(\rho_1)$ be defined by (79). Then:*

- **(Correlation alignment)** *There exist $k_0 \in \mathbb{N}$ and constants $0 < c_- < c_+ < \infty$ (depending only on ρ_1) such that for all $k \geq k_0$,*

$$c_- k \leq w_k \leq c_+ k, \quad \text{hence} \quad 1 - \rho_k = 2w_k^{-2} = O(k^{-2}).$$

- **(Kernel saturation)** *$\Theta^{(k)}(\rho_1) \rightarrow \Theta_\star(r)$ as $k \rightarrow \infty$, where $\Theta_\star(r)$ is the fixed point of the limiting recursion obtained by setting $\dot{s} = s = 1/2$, namely*

$$\Theta_\star(r) = \frac{2r + 1}{2r - 1}.$$

Moreover, there exists $C_\Theta < \infty$ (depending on r, ρ_1) such that for all $k \geq 1$,

$$|\Theta^{(k)}(\rho_1) - \Theta_\star(r)| \leq \frac{C_\Theta}{k}.$$

- **(Depth-induced gap decay)** *Let $\Theta_{\text{diag}}^{(k)} = \Theta^{(k)}(1)$ and $\Theta_{\text{off}}^{(k)} = \Theta^{(k)}(\rho_1)$. Then*

$$\Theta_{\text{diag}}^{(k)} - \Theta_{\text{off}}^{(k)} = \Theta\left(\frac{1}{r k}\right),$$

i.e. there exist $k_1 \in \mathbb{N}$ and constants $0 < c_\Delta < C_\Delta < \infty$ (depending on r, ρ_1) such that for all $k \geq k_1$,

$$\frac{c_\Delta}{r k} \leq \Theta_{\text{diag}}^{(k)} - \Theta_{\text{off}}^{(k)} \leq \frac{C_\Delta}{r k}.$$

Proof See Appendix D.5. ■

Proposition 8 (Centered eigenvalue in an equicorrelated model) *Fix $L \geq 2$ and $r > 1$. Consider a dataset $\{x_i\}_{i=1}^n$ with $\|x_i\| = 1$ and $\rho_{1,ij} = \rho_0 \in (-1, 1)$ for all $i \neq j$. Let*

$$K_{ij}^{(L)} := \Theta^{(L)}(\rho_{1,ij}), \quad H := I_n - \frac{1}{n} \mathbf{1}\mathbf{1}^\top, \quad K_c^{(L)} := H K^{(L)} H.$$

Then $K^{(L)}$ has the explicit form $K^{(L)} = \Theta^{(L)}(1) I_n + \Theta^{(L)}(\rho_0) (\mathbf{1}\mathbf{1}^\top - I_n)$. In particular, the nonzero eigenvalues of $K_c^{(L)}$ are all equal to

$$\lambda_\perp(K^{(L)}) = \Theta^{(L)}(1) - \Theta^{(L)}(\rho_0),$$

with multiplicity $n - 1$. Moreover, by Theorem 7, there exist constants $0 < c_\Delta < C_\Delta < \infty$ (depending on r, ρ_0) such that for all sufficiently large L ,

$$\frac{c_\Delta}{rL} \leq \lambda_\perp(K^{(L)}) \leq \frac{C_\Delta}{rL}.$$

Proof See Appendix D.6. ■

Proposition 9 (Main result: uncentered spectrum in the equicorrelated model) *Under the assumptions of Proposition 8, the eigenvalues of the uncentered Gram matrix $K^{(L)}$ are explicit:*

$$\lambda_1(K^{(L)}) = \Theta^{(L)}(1) + (n-1)\Theta^{(L)}(\rho_0), \quad \lambda_2(K^{(L)}) = \dots = \lambda_n(K^{(L)}) = \Theta^{(L)}(1) - \Theta^{(L)}(\rho_0).$$

In particular, $\lambda_1(K^{(L)})$ corresponds to the constant mode aligned with $\mathbf{1}$, while the spectrum on $\mathbf{1}^\perp$ equals $\lambda_\perp(K^{(L)})$. Moreover, as $L \rightarrow \infty$ (with r fixed) this constant-mode eigenvalue saturates:

$$\lambda_1(K^{(L)}) = n \Theta_\star(r) + O\left(\frac{n}{L}\right), \quad \Theta_\star(r) = \frac{2r+1}{2r-1} = 1 + O\left(\frac{1}{r}\right),$$

and $\lambda_\perp(K^{(L)}) = \Theta(1/(rL))$ by Proposition 8. Thus, in this model the depth/rank scaling is

$$\lambda_{\max}(K^{(L)}) \asymp n \left(1 + \frac{1}{r}\right), \quad \lambda_{\min}(HK^{(L)}H) \asymp \frac{1}{rL}.$$

Proof See Appendix D.7. ■

Remark 10 (Conditioning and time scales in the equicorrelated model) *In the equicorrelated model, the (uncentered) Gram matrix $K^{(L)}$ has eigenvalues $\lambda_1(K^{(L)}) \asymp n$ and $\lambda_\perp(K^{(L)}) \asymp 1/(rL)$ (Proposition 9). For the $1/n$ -normalized square loss, the relevant operator is $(1/n)K^{(L)}$, whose largest eigenvalue is $\lambda_1(K^{(L)})/n = \Theta(1)$ while its smallest eigenvalue is $\lambda_\perp(K^{(L)})/n = \Theta(1/(nrL))$. Therefore the condition number of $(1/n)K^{(L)}$ scales as*

$$\kappa\left(\frac{1}{n}K^{(L)}\right) = \frac{\lambda_1(K^{(L)})/n}{\lambda_\perp(K^{(L)})/n} = \Theta(nrL).$$

Under the parameter budget scaling $P \asymp LrN$ and the NTK width scaling $N \asymp n^2$, this becomes $\kappa((1/n)K^{(L)}) = \Theta(P/n)$.

If the targets are mean-zero (or labels are centered), the constant mode decouples and one may restrict to $\mathbf{1}^\perp$, where in the equicorrelated model $(1/n)HK^{(L)}H$ has flat spectrum (condition number 1) with eigenvalue $\lambda_\perp(K^{(L)})/n = \Theta(1/(nrL))$. Equivalently, the optimization time scale is $\Theta(n/\lambda_\perp(K^{(L)})) = \Theta(nrL)$, which becomes $\Theta(P/n)$ under $P \asymp LrN$ and $N \asymp n^2$.

Comparison to fully trained MLPs at the edge of chaos. For fully trained MLPs at the EOC, dataset-level spectral control is commonly phrased in terms of the centered operator $(1/n)HKH$, and inverse-cosine-distance matrix bounds yield worst-case lower bounds on the smallest nontrivial eigenvalue of order $1/n$ under generic non-degeneracy assumptions on the dataset [Terjék and González-Sánchez \(2025\)](#). Since $\lambda_{\max}((1/n)HKH) = O(1)$ for bounded kernels, these bounds correspond to a condition number scaling $\kappa((1/n)HKH) = \Theta(n)$. In contrast, in the RF-LR equicorrelated model, the full operator $(1/n)K^{(L)}$ has condition number $\Theta(P/n)$, while on $\mathbf{1}^\perp$ the spectrum is flat and the slow time scale is $\Theta(P/n)$.

Corollary 11 (Optimization time scale and P/n law in the equicorrelated model) Under the assumptions of Proposition 8, consider kernel gradient flow for the $1/n$ -normalized square loss with centered targets (equivalently, restrict to $\mathbf{1}^\perp$). Then the operator $(1/n)HK^{(L)}H$ has nonzero eigenvalue $\lambda_\perp(K^{(L)})/n$ with multiplicity $n - 1$, hence the natural optimization time scale is set by $n/\lambda_\perp(K^{(L)})$. Moreover, for large L one has

$$\frac{\lambda_\perp(K^{(L)})}{n} = \Theta\left(\frac{1}{nrL}\right)$$

by Proposition 8. If additionally the per-layer feature widths satisfy $n_\ell \asymp N$ and the trainable parameter budget scales as $P \asymp LrN$, then

$$\frac{n}{\lambda_\perp(K^{(L)})} = \Theta\left(\frac{nP}{N}\right).$$

Under the standard kernel-regime scaling $N \asymp n^2$, this simplifies to

$$\frac{n}{\lambda_\perp(K^{(L)})} = \Theta\left(\frac{P}{n}\right).$$

Technical remarks (appendix). For any finite network, the centered empirical Gram matrix $H\hat{K}H$ is positive semidefinite, and the mean-kernel infinite-depth limit collapses after centering; see Appendix D.8 and Appendix D.9.

5. Low rank is enough for the NTK RKHS (three-layer mean kernel)

We show that low rank does not shrink the kernel-regime function class: the mean three-layer RF-LR kernel induces the same RKHS as the shallow ReLU kernel.

5.1. Microscopic distributions: Fisher–Kibble decoupling

We can now state a compact representation under a homogeneity assumption.

Lemma 12 (Fisher–Kibble decoupling) Let x, y be input vectors and let x_1, y_1 denote their rank- r random projections. Define the empirical correlation $\rho_1 = \cos \angle(x_1, y_1)$ and squared norms $u = \|x_1\|^2, v = \|y_1\|^2$. Then:

- ρ_1 follows Fisher’s correlation distribution [Fisher \(1915\)](#); [Hotelling \(1953\)](#), centered at the true correlation ρ .

- (u, v) follow Kibble's [Kibble \(1941\)](#) bivariate Gamma (chi-square) law.
- Angular and radial parts are independent: $p(\rho_1, u, v) = p_{\text{Fisher}}(\rho_1) p_{\text{Kibble}}(u, v)$.

Proof: See Appendix [C.2](#), Proof [C.2](#).

Homogeneity assumption enabling decoupling. By positive 1-homogeneity and rotational invariance, the base kernel factorizes and $(\rho_1, \|x_1\|, \|y_1\|)$ decouple into angular (Fisher) and radial (Kibble) components; see Appendix [22](#) and Appendix [C.2](#) for details, explicit densities, and the induced compact three-layer empirical NTK form.

5.2. Endpoint expansion via hypergeometric analysis

The RKHS of zonal kernels on the sphere is controlled by the Puiseux behavior near $\rho = \pm 1$ [Bietti and Bach \(2021\)](#). For the mean three-layer kernel, the key step is to analyze the Fisher expectation $\mathbb{E}[\arccos(\hat{\rho}_r)]$ as $\rho \rightarrow 1$.

Proposition 13 (Puiseux expansion of $\mathbb{E}(\arccos(\hat{\rho}_r))$ near $\rho = 1$) *Let $r > 2$ and $\hat{\rho}_r \sim \text{Fisher}(\rho, r)$. Write $\rho = 1 - t$ with $t \downarrow 0$. Then*

$$\mathbb{E}[\arccos(\hat{\rho}_r)] = \sqrt{2t} I(r) + O(t^{3/2}),$$

where $I(r) \in (0, \infty)$ admits the closed form

$$I(r) = \frac{(r-2) 2^{r-\frac{5}{2}}}{\sqrt{2\pi}} \frac{\Gamma(\frac{r-1}{2}) \Gamma(\frac{r}{2} - 1)}{\Gamma(r-1)} C_1(r), \quad C_1(r) = \frac{\Gamma(r - \frac{1}{2}) \Gamma(r - \frac{3}{2})}{\Gamma(r-1)^2}. \quad (8)$$

Proof [Proof sketch] Insert Fisher's density (Appendix [22](#)) into $\int_{-1}^1 \arccos(s) p_{\text{Fisher}}(s \mid 1-t, r) ds$ and change variables $s = 1 - v$. Near $t, v \rightarrow 0$, use $\arccos(1-v) = \sqrt{2v} + O(v^{3/2})$ and apply the hypergeometric connection formula (DLMF §15.8.2) to replace ${}_2F_1(\frac{1}{2}, \frac{1}{2}; r - \frac{1}{2}; 1 - \frac{t+v}{2})$ by the constant $C_1(r)$ up to $O((t+v)^{r-3/2})$. Then the leading integral reduces, after $w = v/t$, to a Beta integral and yields (8). See Appendix [C.6](#) for the full proof. ■

Proposition 14 ($I(r)$ decays as $1/\sqrt{r}$) *Let $I(r)$ be defined by (8). Then $I(r) = O(1/\sqrt{r})$ as $r \rightarrow \infty$.*

Proof [Proof sketch] Apply Stirling's formula to the Gamma ratios in (8). A detailed proof is given in Proposition [29](#) (Appendix [C.7](#)). ■

5.3. RKHS equivalence via endpoint behavior

We now state the RKHS characterization result used to translate Puiseux endpoint behavior into an RKHS statement.

Theorem 15 (Bietti–Bach RKHS criterion (informal)) *Let K_1, K_2 be zonal kernels on $\mathbb{S}^{d-1}$ that are C^∞ on $[-1, 1)$. Suppose that as $t \downarrow 0$,*

$$K_i(1 - t) = K_i(1) - c_i t^\alpha + O(t^{\alpha+\eta}), \quad c_i > 0,$$

for the same exponent $\alpha \in (0, 1)$ and some $\eta > 0$ (and likewise at $\rho = -1$ if needed). Then K_1 and K_2 induce the same RKHS on $\mathbb{S}^{d-1}$, with equivalent norms.

Proof This is a direct consequence of (Bietti and Bach, 2021, Theorem 1), which relates the decay of spherical harmonic coefficients (hence the RKHS) to the Puiseux exponent at the endpoints. ■

Corollary 16 (Low rank is enough for the NTK RKHS: three-layer mean kernel) *Under the RF-LR setting with ReLU nonlinearity and isotropic random features on $\mathbb{S}^{d-1}$, the three-layer mean NTK $\tilde{\Theta}^{(2)}$ induces the same RKHS as the shallow ReLU kernel. In particular, the corresponding RKHSs coincide as sets with equivalent norms.*

Proof [Proof sketch] The mean three-layer kernel is zonal. Proposition 13 gives a Puiseux expansion near $\rho = 1$ with leading exponent $t^{1/2}$; Proposition 14 shows the r -dependent coefficient is an $O(1)$ perturbation of the shallow ReLU coefficient. Since the shallow ReLU kernel has the same endpoint exponent $1/2$, Theorem 15 yields RKHS equivalence. See Appendix C.6 for a detailed argument. ■

Open problem: depth $L \geq 3$. Extending RKHS equivalence to general depth requires controlling endpoint expansions through the full Fisher chain; we leave this to future work (Appendix C.6).

Concentration (appendix). We also prove that the empirical three-layer NTK concentrates around its deterministic limit with sub-Gaussian tails in r ; see Corollary 25 and Appendix C.5. The $I(r) = O(1/\sqrt{r})$ scaling (Proposition 29) is consistent with the fact that Fisher–Kibble corrections to the mean kernel’s leading endpoint coefficient vanish as r grows.

6. Conclusion

This work develops an NTK analysis for low-rank random-feature architectures (RF-LR/MMNN) under explicit assumptions. **Expressivity (RKHS; secondary consequence).** In the NTK regime, the three-layer RF-LR mean kernel induces the same RKHS as the shallow ReLU kernel (Corollary 16), so depth does not enlarge the RKHS in this setting. **Concentration in the bottleneck dimension.** For the three-layer case, the empirical kernel concentrates around its deterministic limit $K_\infty(\rho)$ with tails controlled by the bottleneck dimension r (Theorem 24, Corollary 25). **Centered conditioning and optimization scales.** In an equicorrelated model, the centered Gram matrix HKH has flat spectrum on $\mathbf{1}^\perp$ (condition number 1), and its nontrivial eigenvalue scale is $\lambda_\perp = \Theta(1/(rL))$ (Section 4), yielding the budget-driven kernel time scale $\Theta(P/n)$ under $P \asymp LrN$ and $N \asymp n^2$. **Recursion.** We provide an explicit NTK recursion for arbitrary depth in RF-LR and a closed-form expansion (Theorem 4, Corollary 5), which can be used as a starting point for finite-width corrections.

7. Discussion

This work leaves several natural directions open. We emphasize that our current scope is the lazy/NTK regime under explicit standard assumptions; extending beyond this setting requires additional ideas.

A recurring qualitative message, shared with depth analyses for fully trained MLPs at the edge of chaos [Terjék and González-Sánchez \(2025\)](#); [Hayou et al. \(2022\)](#), is that increasing depth tends to drive correlations toward 1, so layerwise kernels become nearly constant and the deep kernel approaches a rank-one structure. In RF-LR, this effect is compounded by the explicit $1/r$ bottleneck factors in the recursion, which geometrically suppress early-layer contributions (Section 4); consequently, after centering the relevant spectrum on $1^\perp$ collapses unless one rescales time/learning rate (Remark 35 in Appendix D.9).

A technical feature of our setup is that freezing the random features removes some cross-layer dependencies that appear in fully trained networks. Combined with Fisher–Kibble decoupling [Fisher \(1915\)](#); [Hotelling \(1953\)](#), this makes certain calculations more explicit and suggests a path toward extending the analysis to joint large- n regimes using resolvent replacement and Gaussian-equivalence methods developed for shallow two-layer NTKs [Benigni and Paquette \(2025\)](#).

The recursion (Theorem 4) suggests organizing finite-width corrections as an expansion in the bottleneck dimension r . Making this program rigorous for general depth would require controlling the growth of cross-terms produced by the recursion and propagating concentration through the composed Gaussian-process layers.

On the theoretical front, several important questions remain open and constitute an immediate research agenda. Extending the exponential concentration bounds to general depth L layers requires controlling nested conditional means through L layers and managing a quadratic number $O(L^2)$ of cross-terms arising from the recursive NTK structure. The technical challenge lies in tracking how the random correlations ρ_ℓ and norm products $\|h^{(\ell)}(x)\| \|h^{(\ell)}(x')\|$ propagate through layers while maintaining exponential concentration rates. A **non-Gaussian process propagation analysis** distinct from NNGP fits would provide complementary insights beyond the kernel regime, potentially revealing how feature learning emerges from deviations from Gaussianity. An important future direction is to establish **bulk spectral laws** for the empirical RF-LR NTK Gram matrix in a quadratic scaling regime $n \rightarrow \infty$, adapting the Benigni–Paquette framework beyond shallow two-layer kernels to the present low-rank random-feature structure and to depth $L \geq 2$ [Benigni and Paquette \(2025\)](#).

Additional theoretical extensions hold particular promise. The current analysis focuses on the three-layer case for tractability; extending to arbitrary depth L while maintaining explicit formulas remains an open challenge. Investigating how the edge-of-chaos (EOC) parameterization affects finite-width corrections and the Nr scaling mentioned in the architecture definition would provide concrete quantitative bounds. Understanding how the curse of dimensionality impacts normalization choices (unit norm weights vs. biases) depending on data geometry (hypercube vs. spherical) would provide principled guidance for practitioners.

The experimental validation of our theoretical predictions constitutes an equally important and concrete next step. We envision a comprehensive program that includes **confirming lazy-kernel predictions** and precisely mapping the boundary of lazy training in practice; conducting systematic **three-layer width sweeps** to test the $O(1/n) + O(1/r)$ finite-width corrections suggested by the expansion and identify minimal (n, r) configurations where kernel predictions become accurate;

and a detailed **comparison suite with MLPs** that probes centered conditioning and optimization time scales at matched parameter counts. These studies would verify the **exponential concentration** rates (Theorem 24) and demonstrate the **practical accessibility of the kernel regime** for low-rank random features.

Finally, it would be valuable to connect the low-rank random-feature recursion studied here with broader frameworks for infinite-width limits (e.g. tensor programs Yang (2020)) and with depth-dependent NTK spectral analyses for MLPs at the edge of chaos Terjék and González-Sánchez (2025), while maintaining explicit control of the bottleneck scaling.

Our present scope is deliberately focused on the **lazy/NTK regime**, where analytical tractability is greatest and the advantages of low-rank structure are most clearly visible. Completing finite-width formulas, spectrum characterizations, and experimental validations would strengthen the connection between the asymptotic theory and practical regimes.

References

- Sanjeev Arora, Nadav Cohen, and Elad Hazan. On the optimization of deep networks: Implicit acceleration by overparameterization. *International Conference on Machine Learning*, pages 244–253, 2018.
- Sanjeev Arora, Simon S Du, Wei Hu, Zhiyuan Li, and Ruosong Wang. On exact computation with an infinitely wide neural net. *Advances in Neural Information Processing Systems*, 32, 2019.
- Lucas Benigni and Elliot Paquette. Eigenvalue distribution of the neural tangent kernel in the quadratic scaling, 2025. URL <https://arxiv.org/abs/2508.20036>.
- Alberto Bietti and Francis Bach. Deep equals shallow for relu networks in kernel regimes, 2021. URL <https://arxiv.org/abs/2009.14397>. ICLR 2021.
- Yuan Cao, Zhiying Fang, Yue Wu, Ding-Xuan Zhou, and Quanquan Gu. Towards understanding the spectral bias of deep learning, 2020. URL <https://arxiv.org/abs/1912.01198>.
- Lenaïc Chizat and Francis Bach. On the global convergence of gradient descent for overparameterized models using optimal transport. *Advances in Neural Information Processing Systems*, 31, 2018.
- Jeremy M. Cohen, Alex Damian, Ameet Talwalkar, J. Zico Kolter, and Jason D. Lee. Understanding optimization in deep learning with central flows, 2025. URL <https://arxiv.org/abs/2410.24206>.
- Arif Kerem Dayi and Sitan Chen. Gradient dynamics for low-rank fine-tuning beyond kernels, 2024. URL <https://arxiv.org/abs/2411.15385>.
- Ethan Dyer and Guy Gur-Ari. Asymptotics of wide networks from feynman diagrams, 2019. URL <https://arxiv.org/abs/1909.11304>.
- Weinan E, Chao Ma, and Lei Wu. The barron space and the flow-induced function spaces for neural network models, 2021. URL <https://arxiv.org/abs/1906.08039>.

- Ronald Aylmer Fisher. On the "probable error" of a coefficient of correlation deduced from a small sample. *Metron*, 1:3–32, 1915.
- Max Guillen, Philipp Misof, and Jan E. Gerken. Finite-width neural tangent kernels from feynman diagrams, 2025. URL <https://arxiv.org/abs/2508.11522>.
- Soufiane Hayou, Arnaud Doucet, and Judith Rousseau. On the selection of initialization and activation function for deep neural networks, 2018. URL <https://arxiv.org/abs/1805.08266>.
- Soufiane Hayou, Arnaud Doucet, and Judith Rousseau. The curse of depth in kernel regime. In Melanie F. Pradier, Aaron Schein, Stephanie Hyland, Francisco J. R. Ruiz, and Jessica Z. Forde, editors, *Proceedings on "I (Still) Can't Believe It's Not Better!" at NeurIPS 2021 Workshops*, volume 163 of *Proceedings of Machine Learning Research*, pages 41–47. PMLR, 13 Dec 2022. URL <https://proceedings.mlr.press/v163/hayou22a.html>.
- Harold Hotelling. New light on the correlation coefficient and its transforms. *Journal of the Royal Statistical Society: Series B (Methodological)*, 15(2):193–225, 1953. ISSN 0035-9246. doi: 10.1111/j.2517-6161.1953.tb00135.x. URL <https://doi.org/10.1111/j.2517-6161.1953.tb00135.x>.
- Jiaoyang Huang and Horng-Tzer Yau. Dynamics of deep neural networks and neural tangent hierarchy, 2019. URL <https://arxiv.org/abs/1909.08156>.
- Arthur Jacot, Franck Gabriel, and Clément Hongler. Neural tangent kernel: Convergence and generalization in neural networks. In *Advances in Neural Information Processing Systems*, pages 8571–8580, 2018.
- Uijeong Jang, Jason D. Lee, and Ernest K. Ryu. Lora training in the ntk regime has no spurious local minima, 2024. URL <https://arxiv.org/abs/2402.11867>.
- Siddhartha Rao Kamalakara, Acyr Locatelli, Bharat Venkitesh, Jimmy Ba, Yarin Gal, and Aidan N. Gomez. Exploring low rank training of deep neural networks, 2022. URL <https://arxiv.org/abs/2209.13569>.
- W. F. Kibble. A two-variate gamma type distribution. *Sankhyā: The Indian Journal of Statistics*, 5:137–150, 1941. URL <https://www.researchgate.net/profile/Hayat-Rezgui/post/Coefficient-estimates-for-fitting-data-to-a-gamma-variate-function/attachment/59d6550479197b80779ac588/AS%3A523656683102208%401501860890277/download/A+Two-Variate+Gamma+Type+Distribution.pdf>.
- B. Laurent and P. Massart. Adaptive estimation of a quadratic functional by model selection. *The Annals of Statistics*, 28(5):1302 – 1338, 2000. doi: 10.1214/aos/1015957395. URL <https://doi.org/10.1214/aos/1015957395>.
- Song Mei, Andrea Montanari, and Phan-Minh Nguyen. A mean field view of the landscape of two-layer neural networks. *Proceedings of the National Academy of Sciences*, 115(33):E7665–E7671, 2018.

- Song Mei, Theodor Misiakiewicz, and Andrea Montanari. Mean field theory of two-layers neural networks: dimension-free bounds and kernel limit. *Conference on Learning Theory*, pages 2388–2464, 2019.
- Sanghoon Na and Haizhao Yang. Curse of dimensionality in neural network optimization, 2025. URL <https://arxiv.org/abs/2502.05360>.
- Quynh Nguyen, Marco Mondelli, and Guido Montufar. Tight bounds on the smallest eigenvalue of the neural tangent kernel for deep relu networks, 2022. URL <https://arxiv.org/abs/2012.11654>.
- Alexander Novikov, Dmitry Podoprikin, Anton Osokin, and Dmitry Vetrov. Tensorizing neural networks. In *Advances in Neural Information Processing Systems*, pages 442–450, 2015.
- F. W. J. Olver, A. B. Olde Daalhuis, D. W. Lozier, B. I. Schneider, R. F. Boisvert, C. W. Clark, B. R. Miller, B. V. Saunders, H. S. Cohl, and M. A. McClain. Nist digital library of mathematical functions. <https://dlmf.nist.gov/>, 2024. Release 1.2.1 of 2024-06-15.
- Maziar Raissi, Paris Perdikaris, and George Em Karniadakis. Physics informed deep learning (part i): Data-driven solutions of nonlinear partial differential equations, 2017. URL <https://arxiv.org/abs/1711.10561>.
- Grant M Rotskoff and Eric Vanden-Eijnden. Neural networks as interacting particle systems: Asymptotic convexity of the loss landscape and universal scaling of the approximation error. *arXiv preprint arXiv:1805.00915*, 2018.
- Tara N Sainath, Brian Kingsbury, Vikas Sindhwani, Ebru Arisoy, and Bhuvana Ramabhadran. Low-rank matrix factorization for deep neural network training with high-dimensional output targets. *IEEE International Conference on Acoustics, Speech and Signal Processing*, pages 6655–6659, 2013.
- Milad Sefidgaran, Kimia Nadjahi, and Abdellatif Zaidi. Tighter cmi-based generalization bounds via stochastic projection and quantization, 2025. URL <https://arxiv.org/abs/2510.23485>.
- Dávid Terjék. The spectrum of the neural tangent kernel at initialization. *arXiv preprint arXiv:2211.10688*, 2022.
- Dávid Terjék and Diego González-Sánchez. Mlps at the eoc: Spectrum of the ntk. *JMLR*, 23, 2025. Rényi Institute.
- Roman Vershynin. *High-Dimensional Probability: An Introduction with Applications in Data Science*. Cambridge University Press, 2018. doi: 10.1017/9781108231596.
- Greg Yang. Scaling limits of wide neural networks with weight sharing: Gaussian process behavior, gradient independence, and neural tangent kernel derivation. *arXiv preprint arXiv:1902.04760*, 2019.
- Greg Yang. Tensor programs ii: Neural tangent kernel for any architecture, 2020. URL <https://arxiv.org/abs/2006.14548>.

Greg Yang, Dingli Yu, Chen Zhu, and Soufiane Hayou. Tensor programs vi: Feature learning in infinite-depth neural networks, 2023. URL <https://arxiv.org/abs/2310.02244>.

Shijun Zhang, Hongkai Zhao, Yimin Zhong, and Haomin Zhou. Structured and balanced multi-component and multi-layer neural networks, 2025. URL <https://arxiv.org/abs/2407.00765>.

Appendix A. Notation and standing assumptions

A.1. Standing assumptions (expanded)

We summarize the conventions used throughout the main text.

- **Sequential infinite width.** Widths $n_1, \dots, n_L$ tend to infinity sequentially under NTK scaling $1/\sqrt{n_\ell}$. Random features $w^{(\ell)}$ are i.i.d. Gaussian and frozen after initialization.
- **Training policy.** Only the readouts $A^{(\ell)}$ (and layer biases $c^{(\ell)}$) are trained; all other parameters are frozen.
- **Bottleneck regime.** For $\ell \geq 2$, the representation dimension is $d_\ell = r \ll n_\ell$. This induces an explicit $1/r$ factor in the RF-LR NTK recursion at bottleneck layers.
- **Zonal kernels.** By rotational invariance of Gaussian features, base and derivative kernels depend on inputs only through the cosine similarity $\rho = \langle x, x' \rangle / (\|x\| \|x'\|)$. For RKHS results we typically take $x \in \mathbb{S}^{d-1}$.
- **Homogeneity.** We use Assumption 1 (positive 1-homogeneity) for Fisher–Kibble decoupling and for separating radial and angular components in three-layer calculations.

Assumption 1 (Homogeneous activation) *The activation function σ is positively 1-homogeneous:*

$$\sigma(\alpha u) = \alpha \sigma(u) \quad \text{for all } \alpha \geq 0, u \in \mathbb{R}.$$

In particular, σ satisfies this property. This assumption is used for Fisher–Kibble decoupling (Lemma 12) and for separating radial and angular fluctuations in the three-layer kernel.

A.2. EOC scaling and the origin of $1/r$

Let $\Sigma_w = \text{Cov}(w_j^{(\ell)})$. Writing $q^{(\ell)} = \mathbb{E} \|h^{(\ell)}(x)\|^2$ for a fixed unit-norm input, weight independence yields the variance recursion

$$q^{(\ell)} = \frac{\sigma_A^2}{2} q^{(\ell-1)} \text{Tr}(\Sigma_w),$$

The edge-of-chaos (EOC) condition chooses σ_A so that the linearized coefficient equals 1,

$$\frac{\sigma_A^2}{2} \text{Tr}(\Sigma_w) = 1,$$

preventing exploding/vanishing signal propagation across depth. In particular, under $\Sigma_w = I_{d_{\ell-1}}/d_{\ell-1}$ one has $\text{Tr}(\Sigma_w) = 1$ and $\sigma_A = \sqrt{2}$.

In the bottleneck regime $d_{\ell-1} = r$ for $\ell \geq 2$, EOC prescribes $\Sigma_w = I_r/r$. For ReLU, the (normalized) derivative kernel has the closed form

$$\bar{\Sigma}^{(\ell)}(\rho) = \mathbb{P}(Z > 0, Z' > 0) = \frac{1}{2} - \frac{\arccos(\rho)}{2\pi} \in \left[0, \frac{1}{2}\right],$$

so $\dot{\Sigma}^{(\ell)}(x, x') = \bar{\Sigma}^{(\ell)}(\rho_\ell)$ remains $O(1)$ under EOC and is uniformly bounded by $1/2$. The RF-LR NTK recursion (Theorem 4) carries an explicit prefactor $1/r$ at bottleneck layers because gradients propagate through an r -dimensional bottleneck and the corresponding Jacobian metric concentrates on an r -dimensional subspace; see Appendix B.1 for the detailed derivation.

A.3. Comparison with the classical fully-trained NTK recursion

For comparison, consider a standard fully-connected MLP where *all* weights (and biases) are trained under NTK parameterization. In the sequential infinite-width limit, the classical NTK recursion takes the form [Jacot et al. \(2018\)](#)

$$\Theta_{\text{MLP}}^{(0)}(x, x') = 0, \quad \Theta_{\text{MLP}}^{(\ell)}(x, x') = \Theta_{\text{MLP}}^{(\ell-1)}(x, x') \dot{K}^{(\ell)}(x, x') + K^{(\ell)}(x, x'), \quad \ell \geq 1, \quad (9)$$

where $K^{(\ell)}$ is the (deterministic) NNGP kernel at layer ℓ and $\dot{K}^{(\ell)}$ the associated derivative kernel (both obtained from the signal-propagation recursion). Unrolling (9) yields one contribution per layer (a sum of L terms, each propagated through subsequent $\dot{K}^{(k)}$ factors).

In contrast, the RF-LR recursion of Theorem 4 (i) carries explicit $1/r$ prefactors at bottleneck layers, and (ii) uses conditional base/derivative kernels $\Sigma^{(\ell)}, \dot{\Sigma}^{(\ell)}$ rather than the classical NNGP kernels $K^{(\ell)}, \dot{K}^{(\ell)}$. These differences lead to the $2L$ -term expansion in Corollary 5, reflecting separate bias-path and fresh-basis contributions, each propagated through subsequent derivative kernels and weighted by bottleneck factors.

Relation to fully trained MLPs at the edge of chaos. For fully trained (full-width) MLPs at the edge of chaos, the limiting NTK admits a single-layer sum structure (one contribution per layer) without RF-LR bottleneck prefactors; see Proposition 19 for a reference closed form [Terjék and González-Sánchez \(2025\)](#). In contrast, the RF-LR expansion (7) contains $2L$ terms because the recursion has both bias-path and fresh-basis-path contributions, each propagated through subsequent derivative kernels and weighted by explicit $1/r$ factors at bottlenecks.

A.4. Probabilistic recursion, GP concatenation, and bottleneck scaling

Kernel randomness and GP concatenation. For $\ell > 1$, $\Sigma^{(\ell)}$ and $\dot{\Sigma}^{(\ell)}$ are random fields because they depend on $\mathbf{h}^{(\ell-1)}$, which is the output of a (conditional) Gaussian process. By Theorem 3, the hidden states form a composition of GPs: each layer’s output is conditionally Gaussian given the previous layer, but the unconditional law is a deep GP. In the bottleneck regime, the pair $(\mathbf{h}^{(\ell-1)}(x), \mathbf{h}^{(\ell-1)}(x'))$ is a $2r$ -dimensional Gaussian conditional on lower-layer randomness, with covariance determined by $\Sigma^{(\ell-1)}$. The base kernel $\Sigma^{(\ell)}$ and derivative kernel $\dot{\Sigma}^{(\ell)}$ are thus expectations over the frozen features $\mathbf{w}^{(\ell)}$, conditional on this Gaussian pair, and inherit randomness through $\Sigma^{(\ell-1)}$.

Bottleneck scaling and explicit ReLU bounds. At a bottleneck layer $\ell \geq 2$, the RF-LR NTK recursion carries an explicit prefactor $1/r$ (Theorem 4). For ReLU and isotropic weights with $\text{Cov}(w) = I_r/r$, one has

$$\dot{\Sigma}^{(\ell)}(x, x') = \mathbb{E}_w \left[\dot{\sigma}(w^\top \mathbf{h}^{(\ell-1)}(x)) \dot{\sigma}(w^\top \mathbf{h}^{(\ell-1)}(x')) \|w\|^2 \right] = \bar{\Sigma}^{(\ell)}(\rho_\ell),$$

where $\rho_\ell = \cos \angle(\mathbf{h}^{(\ell-1)}(x), \mathbf{h}^{(\ell-1)}(x'))$ and $\bar{\Sigma}^{(\ell)}(\rho)$ denotes the corresponding scalar derivative kernel evaluated at cosine similarity $\rho \in [-1, 1]$. Under the EOC normalization $\text{Tr}(\text{Cov}(w)) = 1$, the ReLU derivative kernel has the explicit form

$$\bar{\Sigma}^{(\ell)}(\rho) = \mathbb{P}(Z > 0, Z' > 0) = \frac{1}{2} - \frac{\arccos(\rho)}{2\pi} \in \left[0, \frac{1}{2}\right],$$

where (Z, Z') is centered Gaussian with correlation ρ . In particular, $\sup_{\rho \in [-1, 1]} \bar{\Sigma}^{(\ell)}(\rho) = 1/2$, uniformly in ℓ . Similarly, $\Sigma^{(\ell)}(x, x') = \mathbb{E}_w[\sigma(w^\top \mathbf{h}^{(\ell-1)}(x)) \sigma(w^\top \mathbf{h}^{(\ell-1)}(x'))]$ is an $O(1)$ scalar kernel, and the $1/r$ factors enter through the recursion.

Probabilistic recursion (nested expectations). The mean (probabilistic) NTK $\tilde{\Theta}^{(L)} = \mathbb{E}[\Theta^{(L)}]$ satisfies a recursion involving expectations of products:

$$\tilde{\Theta}^{(1)} = 1 + \mathbb{E}[\Sigma^{(1)}], \quad \tilde{\Theta}^{(\ell)} = 1 + \frac{1}{r} \mathbb{E}[\Theta^{(\ell-1)} \cdot \dot{\Sigma}^{(\ell)}] + \frac{1}{r} \mathbb{E}[\Sigma^{(\ell)}], \quad \ell \geq 2. \quad (10)$$

To emphasize the nested structure, write $\mathbb{E}_{\leq \ell-1}$ for expectation over all randomness up to depth $\ell-1$, and $\mathbb{E}_{w^{(\ell)}}[\cdot | \mathbf{h}^{(\ell-1)}]$ for expectation over the frozen feature direction $w^{(\ell)}$ at layer ℓ , conditional on the previous hidden states. Then the mean kernels are iterated conditional expectations:

$$\mathbb{E}[\Sigma^{(\ell)}(x, x')] = \mathbb{E}_{\leq \ell-1} \left[\mathbb{E}_{w^{(\ell)}} [\sigma(w^\top \mathbf{h}^{(\ell-1)}(x)) \sigma(w^\top \mathbf{h}^{(\ell-1)}(x')) | \mathbf{h}^{(\ell-1)}] \right],$$

$$\mathbb{E}[\Theta^{(\ell-1)}(x, x') \dot{\Sigma}^{(\ell)}(x, x')] = \mathbb{E}_{\leq \ell-1} \left[\Theta^{(\ell-1)}(x, x') \cdot \mathbb{E}_{w^{(\ell)}} [\dot{\sigma}(w^\top \mathbf{h}^{(\ell-1)}(x)) \dot{\sigma}(w^\top \mathbf{h}^{(\ell-1)}(x')) \|w\|^2 | \mathbf{h}^{(\ell-1)}] \right].$$

Since $\Theta^{(\ell-1)}$ and $\Sigma^{(\ell)}, \dot{\Sigma}^{(\ell)}$ share the randomness of $\mathbf{h}^{(\ell-1)}$, one has $\mathbb{E}[\Theta^{(\ell-1)} \cdot \dot{\Sigma}^{(\ell)}] \neq \tilde{\Theta}^{(\ell-1)} \cdot \mathbb{E}[\dot{\Sigma}^{(\ell)}]$ in general; controlling this dependence (and the resulting nested conditional expectations) is the main technical point for a rigorous deep mean-kernel theory.

Exponential depth suppression (ReLU, bottlenecks). Assume for simplicity that all layers $\ell \geq 2$ are bottlenecks with the same rank r . For ReLU under EOC we have the explicit uniform bound

$$\bar{\Sigma}^{(\ell)}(\rho) \leq c_0 := \frac{1}{2}, \quad \forall \rho \in [-1, 1], \forall \ell \geq 2.$$

From the closed form (7), the contribution from layer ℓ to $\Theta^{(L)}$ is multiplied by $\prod_{k=\ell+1}^L \dot{\Sigma}^{(k)}$. Writing $\dot{\Sigma}^{(k)} = \bar{\Sigma}^{(k)}(\rho_k)$, we obtain

$$\frac{1}{r^{L-\ell}} \prod_{k=\ell+1}^L \dot{\Sigma}^{(k)} \leq \frac{1}{r^{L-\ell}} \prod_{k=\ell+1}^L \bar{\Sigma}^{(k)}(\rho_k) \leq \frac{c_0^{L-\ell}}{r^{L-\ell}}.$$

Hence, contributions from early layers ($\ell \ll L$) decay exponentially in $L - \ell$ when $c_0/r < 1$, and the NTK becomes increasingly localized to the top of the network as depth grows.

Proposition 17 (Effective depth decomposition and a log r window) *Assume the uniform derivative-kernel bound $\dot{\Sigma}^{(\ell)}(x, x') \leq c_0$ for all $\ell \geq 2$ and all input pairs (x, x') , and assume that for a given (x, x') one has a uniform base-kernel bound*

$$|\Sigma^{(\ell)}(x, x')| \leq C_\Sigma(x, x'), \quad \forall \ell \in \{1, \dots, L\}.$$

Fix $L \geq 2$, $r > c_0$, and an integer $m \in \{1, \dots, L\}$. Define the top- m truncation $\Theta_{\text{top},m}^{(L)}(x, x')$ by restricting both sums in the explicit expansion (7) to indices $\ell \in \{L - m + 1, \dots, L\}$. Then

$$|\Theta^{(L)}(x, x') - \Theta_{\text{top},m}^{(L)}(x, x')| \leq \frac{1 + C_\Sigma(x, x')}{1 - c_0/r} \left(\frac{c_0}{r}\right)^m.$$

In particular, taking $m = \lceil \alpha \log r \rceil$ (for any $\alpha > 0$) yields a log r effective-depth window: the contribution of layers $\ell \leq L - \lceil \alpha \log r \rceil$ is at most

$$|\Theta^{(L)}(x, x') - \Theta_{\text{top},\lceil \alpha \log r \rceil}^{(L)}(x, x')| \leq \frac{1 + C_\Sigma(x, x')}{1 - c_0/r} \left(\frac{c_0}{r}\right)^{\lceil \alpha \log r \rceil} = o(r^{-p})$$

for every fixed $p > 0$ as $r \rightarrow \infty$.

Proof By definition of $\Theta_{\text{top},m}^{(L)}$, the remainder is the sum of the terms in (7) with $\ell \leq L - m$. For such ℓ , we have $L - \ell \geq m$, and by the derivative-kernel bound $\dot{\Sigma}^{(k)}(x, x') \leq c_0$ we obtain

$$\frac{1}{r^{L-\ell}} \prod_{k=\ell+1}^L \dot{\Sigma}^{(k)}(x, x') \leq \left(\frac{c_0}{r}\right)^{L-\ell}.$$

Moreover, for $\ell \leq L - m \leq L - 1$ the base-kernel term carries the same prefactor $r^{-(L-\ell)}$ in (7), hence

$$\frac{1}{r^{L-\ell}} |\Sigma^{(\ell)}(x, x')| \prod_{k=\ell+1}^L \dot{\Sigma}^{(k)}(x, x') \leq C_\Sigma(x, x') \left(\frac{c_0}{r}\right)^{L-\ell}.$$

Summing these bounds over $\ell \leq L - m$ and using a geometric-series tail gives

$$|\Theta^{(L)}(x, x') - \Theta_{\text{top},m}^{(L)}(x, x')| \leq (1 + C_\Sigma(x, x')) \sum_{j=m}^{\infty} \left(\frac{c_0}{r}\right)^j = \frac{1 + C_\Sigma(x, x')}{1 - c_0/r} \left(\frac{c_0}{r}\right)^m,$$

as claimed. The log r specialization follows by substituting $m = \lceil \alpha \log r \rceil$ and observing that $(c_0/r)^{\lceil \alpha \log r \rceil} = o(r^{-p})$ for every fixed $p > 0$. $\blacksquare$

A.5. Comparison table: depth/rank effects in the kernel regime

A.6. Techniques and proof sketch

We highlight the main proof ingredients and how they combine.

Recursion and closed form. The RF-LR NTK recursion (Theorem 4) follows from a layer-wise decomposition of the gradient inner product into bias, readout, and backpropagated contributions, together with conditional law of large numbers arguments under the sequential infinite-width limit Jacot et al. (2018). Iterating the recursion yields the explicit $2L$ -term expansion (Corollary 5).

	Fully trained MLP at EOC Terjék and González-Sánchez (2025)	
Trainable parameters	all weight layers	read
Key scaling parameter	depth L , activation Δ_ϕ	
Depth effect (kernel magnitude)	uncentered scale can grow with L	mea
Centered scale	controlled by correlation propagation	equicorrelated

Table 1: Comparison of depth/rank effects in the kernel regime. The MLP-at-EOC column summarizes known fully trained behavior; RF-LR has an explicit $1/r$ bottleneck factor in the recursion, leading to saturation and geometric suppression of early-layer contributions.

Microscopic decoupling and concentration. For three layers, the empirical kernel depends on the sample cosine and norm product of rank- r Gaussian projections. Rotational invariance yields Fisher–Kibble decoupling (Lemma 12), separating angular and radial fluctuations [Fisher \(1915\)](#); [Hotelling \(1953\)](#); [Kibble \(1941\)](#). Concentration of Gaussian norms and sample correlations then yields sub-Gaussian deviations in r for the empirical kernel (Theorem 24 and Corollary 25).

RKHS identification via endpoint expansions. The RKHS of zonal kernels on the sphere is controlled by the Puiseux behavior near $\rho = \pm 1$ [Bietti and Bach \(2021\)](#). We compute the mean three-layer kernel’s endpoint expansion by combining Fisher’s density with a hypergeometric connection formula, showing that the leading $t^{1/2}$ exponent is preserved after taking expectations, hence the mean kernel induces the same RKHS as the shallow ReLU kernel (Corollary 16).

Depth scaling and effective depth. Depth dependence is controlled by correlation propagation and by products of derivative kernels. In the bottleneck regime, the RF-LR recursion carries explicit $1/r$ prefactors at each bottleneck layer, and for ReLU one has $\sup_\rho \tilde{\Sigma}^{(\ell)}(\rho) \leq 1/2$, which yields exponential suppression of early-layer contributions as depth increases (Section 4). A refined mean-kernel statement is obtained by analyzing the deterministic proxy recursion (Theorem 7).

Future work: quadratic random-matrix scaling. Establishing bulk spectral laws for the *empirical* RF-LR NTK in a quadratic scaling regime $n \rightarrow \infty$ requires additional resolvent/local-law inputs. While such bulk laws are now available for shallow two-layer NTKs via resolvent replacement and Gaussian-equivalence machinery [Benigni and Paquette \(2025\)](#), adapting these methods to the present low-rank random-feature structure and to depth $L \geq 2$ is left for future work.

Appendix B. Proofs of NTK Recursions

B.1. Proof of Theorem 4: Infinite-width NTK recursion

Proof We prove the recursion by decomposing the NTK at layer ℓ into contributions of the trainable parameters up to layer ℓ , and invoking the Law of Large Numbers (LLN) under the sequential infinite-width limit. Let $f^{(\ell)}$ denote the network output truncated at layer ℓ . For finite width, the (random) NTK reads

$$\Theta_{\text{rand}}^{(\ell)}(x, x') = \underbrace{(\nabla_{\theta^{(\ell-1)}} f^{(\ell)}(x))^\top (\nabla_{\theta^{(\ell-1)}} f^{(\ell)}(x'))}_{\text{Term 1}} + \underbrace{\sum_{j=1}^{n_\ell} \frac{\partial f^{(\ell)}(x)}{\partial A_j^{(\ell)}} \frac{\partial f^{(\ell)}(x')}{\partial A_j^{(\ell)}}}_{\text{Term 2}}. \quad (11)$$

Bias parameters. We include a trainable layer bias $c^{(\ell)}$ (initialized at zero). The gradient feature with respect to $c^{(\ell)}$ is constant, so it contributes an additive constant term to the NTK; under our normalization this contribution is 1. This yields the additive +1 term in the recursion.

Term 2 (layer- ℓ output weights $A^{(\ell)}$). By the forward pass, $\partial f^{(\ell)}(x)/\partial A_j^{(\ell)} = \frac{\sigma_A}{\sqrt{n_\ell}} \sigma(w_j^{(\ell)\top} \mathbf{h}^{(\ell-1)}(x))$. Therefore

$$\text{Term 2} = \frac{\sigma_A^2}{n_\ell} \sum_{j=1}^{n_\ell} \sigma(w_j^{(\ell)\top} \mathbf{h}^{(\ell-1)}(x)) \sigma(w_j^{(\ell)\top} \mathbf{h}^{(\ell-1)}(x')). \quad (12)$$

Conditionally on $\mathbf{h}^{(\ell-1)}$, the summands are i.i.d. over $w_j^{(\ell)}$. As $n_\ell \rightarrow \infty$ (with lower layers already taken to their limits), LLN yields

$$\text{Term 2} \xrightarrow{p} \sigma_A^2 \Sigma^{(\ell)}(x, x'). \quad (13)$$

Term 1 (previous layers). By the chain rule,

$$\nabla_{\theta^{(\ell-1)}} f^{(\ell)}(x) = \left(\frac{\partial f^{(\ell)}(x)}{\partial \mathbf{h}^{(\ell-1)}(x)} \right) \nabla_{\theta^{(\ell-1)}} \mathbf{h}^{(\ell-1)}(x). \quad (14)$$

Hence Term 1 factorizes as $\Theta_{\text{rand}}^{(\ell-1)}(x, x')$ multiplied by a Jacobian metric:

$$\text{Term 1} = \Theta_{\text{rand}}^{(\ell-1)}(x, x') \cdot \left\langle \frac{\partial f^{(\ell)}}{\partial \mathbf{h}^{(\ell-1)}}(x), \frac{\partial f^{(\ell)}}{\partial \mathbf{h}^{(\ell-1)}}(x') \right\rangle. \quad (15)$$

Taking the inner product and applying LLN as $n_\ell \rightarrow \infty$ yields

$$\left\langle \frac{\partial f^{(\ell)}}{\partial \mathbf{h}^{(\ell-1)}}(x), \frac{\partial f^{(\ell)}}{\partial \mathbf{h}^{(\ell-1)}}(x') \right\rangle \xrightarrow{p} \sigma_A^2 \dot{\Sigma}^{(\ell)}(x, x'). \quad (16)$$

Therefore,

$$\text{Term 1} \xrightarrow{p} \Theta^{(\ell-1)}(x, x') \cdot \sigma_A^2 \dot{\Sigma}^{(\ell)}(x, x'). \quad (17)$$

Combining the bias contribution with the two limits completes the recursion. For bottleneck layers $\ell \geq 2$, the Jacobian and output contributions scale with $1/r$ (from the r -dimensional bottleneck); we absorb σ_A^2 into the normalization. Thus

$$\Theta^{(1)}(x, x') = 1 + \Sigma^{(1)}(x, x'), \quad \Theta^{(\ell)}(x, x') = 1 + \frac{1}{r} \Theta^{(\ell-1)}(x, x') \cdot \sigma_A^2 \dot{\Sigma}^{(\ell)}(x, x') + \frac{1}{r} \sigma_A^2 \Sigma^{(\ell)}(x, x') \quad (\ell \geq 2) \quad (18)$$

The sequential limit assumption ensures that lower-layer random objects have already converged when taking the next-layer limit, justifying the conditional LLN applications. $\blacksquare$

B.2. Proof of Theorem 3: Gaussian process composition

Proof We prove by induction on ℓ that $\mathbf{h}^{(\ell)}$ converges to a Gaussian process with covariance kernel $\Sigma^{(\ell)}$.

Base case $\ell = 0$. For $\ell = 0$, we have $\mathbf{h}^{(0)}(x) = x$, which is deterministic and trivially a Gaussian process.

Inductive step. Assume that $\mathbf{h}^{(\ell-1)}$ converges to a Gaussian process with covariance kernel $\Sigma^{(\ell-1)}$. Consider the forward pass at layer ℓ :

$$\mathbf{h}^{(\ell)}(x) = \frac{1}{\sqrt{n_\ell}} \sum_{j=1}^{n_\ell} A_j^{(\ell)} \sigma\left(w_j^{(\ell)\top} \mathbf{h}^{(\ell-1)}(x)\right) + c^{(\ell)}. \quad (19)$$

For any finite collection of inputs $\{x_1, \dots, x_m\}$, conditionally on $\mathbf{h}^{(\ell-1)}$, the terms $A_j^{(\ell)} \sigma\left(w_j^{(\ell)\top} \mathbf{h}^{(\ell-1)}(x_i)\right)$ are independent across j and identically distributed. Since $A_j^{(\ell)} \sim \mathcal{N}(0, \sigma_A^2)$ and $w_j^{(\ell)}$ are frozen Gaussian draws, the conditional distribution of each term is centered with finite variance (and $c^{(\ell)}$ is deterministic at initialization).

By the multivariate Central Limit Theorem, as $n_\ell \rightarrow \infty$, the vector $(\mathbf{h}^{(\ell)}(x_1), \dots, \mathbf{h}^{(\ell)}(x_m))$ converges in distribution to a multivariate Gaussian. The covariance is given by

$$\mathbb{E}[\mathbf{h}^{(\ell)}(x_i) \mathbf{h}^{(\ell)}(x_j)] = \mathbb{E}_w \left[\sigma\left(w^\top \mathbf{h}^{(\ell-1)}(x_i)\right) \sigma\left(w^\top \mathbf{h}^{(\ell-1)}(x_j)\right) \right] = \Sigma^{(\ell)}(x_i, x_j), \quad (20)$$

where the expectation is taken over the frozen weights w and the limiting Gaussian process $\mathbf{h}^{(\ell-1)}$. The last equality follows from the definition of the base kernel in Definition 2.

The sequential infinite-width limit ensures that $\mathbf{h}^{(\ell-1)}$ has already converged to its Gaussian process limit when taking the limit for layer ℓ , justifying the application of the CLT conditionally on $\mathbf{h}^{(\ell-1)}$. ■

B.3. Proof of three-layer NTK Corollary

Proof Apply Theorem 4 for $\ell = 1$ and $\ell = 2$. For $\ell = 1$, since $\Theta^{(0)} = 0$, the recursion gives:

$$\Theta^{(1)}(x, x') = 1 + \Theta^{(0)}(x, x') \cdot \dot{\Sigma}^{(1)}(x, x') + \Sigma^{(1)}(x, x') = 1 + \Sigma^{(1)}(x, x'). \quad (21)$$

For $\ell = 2$, substitute $\Theta^{(1)} = 1 + \Sigma^{(1)}$ into the recursion:

$$\Theta^{(2)}(x, x') = 1 + \frac{1}{r} \Theta^{(1)}(x, x') \cdot \dot{\Sigma}^{(2)}(x, x') + \frac{1}{r} \Sigma^{(2)}(x, x'). \quad (22)$$

This is the stated three-layer recursion with the bias contribution.

Mean explicit form (low-rank). Under the homogeneous ReLU setting on the sphere, the mean three-layer NTK (with $1/r$ factors) is

$$\tilde{\Theta}^{(2)}(\rho) = 1 + \frac{1}{r} \Theta^{(1)}(\rho) \mathbb{E} \left[1 - \frac{\arccos(\hat{\rho}_r)}{\pi} \right] + \frac{1}{r} \mathbb{E} \left[\Sigma^{(1)}(\hat{\rho}_r) \|x_1\| \|y_1\| \right], \quad (23)$$

where the expectations are over Fisher and Kibble. As $r \rightarrow \infty$, $\hat{\rho}_r \rightarrow \rho$ and $\|x_1\| \|y_1\| / r \rightarrow 1$, so the limit $K_\infty(\rho) = \Theta^{(1)}(\rho)(1 - \arccos(\rho)/\pi) + \Sigma^{(1)}(\rho) + 1$ is recovered (without the $1/r$ prefactors, since the low-rank scaling is absorbed in the limit). ■

B.4. Proof of General L -layer NTK Corollary

Remark 18 (Explicit $2L$ -term unrolling) *Equivalently, expanding the recursion yields an explicit linear number of terms (namely $2L$). For example, for $L = 3$:*

$$\begin{aligned}\Theta^{(3)}(x, x') &= 1 + \frac{1}{r} \dot{\Sigma}^{(2)} \cdot \dot{\Sigma}^{(3)} + \frac{1}{r} \dot{\Sigma}^{(3)} \\ &\quad + \frac{1}{r^2} \Sigma^{(1)} \cdot \dot{\Sigma}^{(2)} \cdot \dot{\Sigma}^{(3)} + \frac{1}{r} \Sigma^{(2)} \cdot \dot{\Sigma}^{(3)} + \frac{1}{r} \Sigma^{(3)}.\end{aligned}\quad (24)$$

Each bottleneck layer contributes a factor $1/r$; layer 1 has no such factor. In total, there are $2L$ terms for depth L .

Proof We prove the explicit form by induction on L , accounting for the $+1$ bias terms in the recursion.

Base case $L = 1$. From Theorem 4, $\Theta^{(1)} = 1 + \Sigma^{(1)}$. The formula with $L = 1$ gives:

$$\Theta^{(1)}(x, x') = \sum_{\ell=1}^1 \left(\prod_{k=\ell+1}^1 \dot{\Sigma}^{(k)}(x, x') \right) + \sum_{\ell=1}^1 \Sigma^{(\ell)}(x, x') \prod_{k=\ell+1}^1 \dot{\Sigma}^{(k)}(x, x'). \quad (25)$$

For the first sum: when $\ell = 1$, $\prod_{k=2}^1 \dot{\Sigma}^{(k)} = 1$ (empty product). For the second sum: when $\ell = 1$, $\Sigma^{(1)} \cdot \prod_{k=2}^1 \dot{\Sigma}^{(k)} = \Sigma^{(1)} \cdot 1 = \Sigma^{(1)}$. Therefore, $\Theta^{(1)} = 1 + \Sigma^{(1)}$, which matches the recursion.

Inductive step. Assume the formula holds for depth $L - 1$:

$$\Theta^{(L-1)}(x, x') = \sum_{\ell=1}^{L-1} \left(\prod_{k=\ell+1}^{L-1} \dot{\Sigma}^{(k)}(x, x') \right) + \sum_{\ell=1}^{L-1} \Sigma^{(\ell)}(x, x') \prod_{k=\ell+1}^{L-1} \dot{\Sigma}^{(k)}(x, x'). \quad (26)$$

By Theorem 4, for $L \geq 2$,

$$\Theta^{(L)}(x, x') = 1 + \frac{1}{r} \Theta^{(L-1)}(x, x') \cdot \dot{\Sigma}^{(L)}(x, x') + \frac{1}{r} \Sigma^{(L)}(x, x'). \quad (27)$$

Substituting the inductive hypothesis and distributing the $1/r$ factor yields the stated closed form; the first sum gains a new term 1 (from $\ell = L$) and each existing term is multiplied by $(1/r) \dot{\Sigma}^{(L)}$, while the second sum gains $(1/r) \Sigma^{(L)}$ and each existing term is multiplied by $(1/r) \dot{\Sigma}^{(L)}$. The first term 1 corresponds to $\ell = L$ in the first sum (with empty product $\prod_{k=L+1}^L \dot{\Sigma}^{(k)} = 1$). The last term $\Sigma^{(L)}$ corresponds to $\ell = L$ in the second sum (with empty product $\prod_{k=L+1}^L \dot{\Sigma}^{(k)} = 1$). Therefore:

$$\Theta^{(L)}(x, x') = \sum_{\ell=1}^L \left(\prod_{k=\ell+1}^L \dot{\Sigma}^{(k)}(x, x') \right) + \sum_{\ell=1}^L \Sigma^{(\ell)}(x, x') \prod_{k=\ell+1}^L \dot{\Sigma}^{(k)}(x, x'), \quad (28)$$

completing the induction. The total number of terms is $L + L = 2L$, which grows linearly with depth. $\blacksquare$

B.5. Reference: full-width MLP limiting NTK at the edge of chaos

Proposition 19 (Closed form for full-width MLP EOC NTK) *Let $l \geq 1$ and consider an infinitely wide l -layer MLP with (a, b) -ReLU activation initialized at the edge of chaos as in [Terjék and González-Sánchez \(2025\)](#). Denote by ϱ the corresponding cosine map and by $\rho_1(x, x')$ the input cosine similarity. Then the limiting NTK satisfies*

$$K_\infty(x, x') = \|x\| \|x'\| \left(\sum_{k=1}^l \varrho^{\circ(k-1)}(\rho_1(x, x')) \prod_{k'=k}^{l-1} \varrho' \left(\varrho^{\circ(k'-1)}(\rho_1(x, x')) \right) \right) I_{m_l},$$

with the convention that the empty product (for $k = l$) equals 1.

Proof This is [Terjék and González-Sánchez \(2025\)](#); we include it for completeness. ■

Appendix C. Proofs of low-rank NTK RKHS

C.1. Background and statements for Section 5

RKHS viewpoint in the NTK regime. In the NTK regime, training dynamics linearize:

$$f_t(x) \approx f_0(x) + \langle \nabla_\theta f_0(x), \theta_t - \theta_0 \rangle.$$

The learned function stays in the RKHS induced by the (limiting) kernel Θ , with norm

$$\|f\|_{\mathcal{H}_\Theta}^2 = \langle f, \Theta^{-1} f \rangle_{L^2}.$$

For RF-LR with frozen random features, universal approximation holds due to full support of the frozen Gaussian draws $w^{(\ell)}$, so any function in the Barron space [E et al. \(2021\)](#) can be approximated with error $O(1/\sqrt{N})$. Classical NTK analysis suggests that feature weights move little during training; RF-LR enforces this by design by freezing the feature weights, which isolates kernel-regime behavior without an additional weight-deviation analysis.

Mean RKHS and random features. Each realization ω of the random features yields a kernel $K_\omega(x, x') = \Theta_\omega^{(2)}(x, x')$ and an induced RKHS $\mathcal{H}_\omega$. The mean kernel $\tilde{K}(x, x') = \mathbb{E}_\omega[K_\omega(x, x')]$ is deterministic; its RKHS $\mathcal{H}_{\tilde{K}}$ is the RKHS of $\tilde{K}$. It is not $\mathbb{E}[\mathcal{H}_\omega]$ (expectation of Hilbert spaces is undefined). For each ω , the feature map $\varphi_\omega(x) = K_\omega(\cdot, x) \in \mathcal{H}_\omega$ is random, whereas the canonical feature map of the mean kernel $\varphi_{\tilde{K}}(x) = \tilde{K}(\cdot, x) \in \mathcal{H}_{\tilde{K}}$ is deterministic. The mean RKHS describes the typical function space obtained after averaging over random feature realizations.

Remark (application of Bietti–Bach). Once the mean kernel’s Puiseux endpoint behavior is established (Appendix C.6), the RKHS equivalence in Corollary 16 follows directly from the characterization theorem of [Bietti and Bach \(2021\)](#).

Corollary 20 (Mean NTK over Fisher–Kibble has same RKHS: three-layer case) *Under Assumption 1 and the RF-LR setting with ReLU nonlinearity and isotropic random features on $\mathbb{S}^{d-1}$, the mean three-layer NTK $\tilde{\Theta}^{(2)}$ (obtained by taking expectations over Fisher and Kibble distributions) is a zonal kernel that induces the same RKHS as the shallow ReLU kernel. In particular, the RKHSs coincide as sets with equivalent norms.*

Proof: See Appendix C.6.

Main RKHS result (stated in the main text). Corollary 16 is stated in Section 5. It follows by combining the Puiseux endpoint analysis in Appendix C.6 with the RKHS characterization of Bietti and Bach (2021).

C.2. Discussion on Fisher and Kibble Distributions

Lemma 21 (Fisher–Kibble decoupling) *Let x, y be input vectors and let x_1, y_1 denote their rank- r random projections. Define the empirical correlation $\rho_1 = \cos \angle(x_1, y_1)$ and squared norms $u = \|x_1\|^2, v = \|y_1\|^2$. Then:*

- ρ_1 follows Fisher’s correlation distribution [Fisher \(1915\)](#) [Hotelling \(1953\)](#), centered at the true correlation ρ .
- (u, v) follow Kibble’s [Kibble \(1941\)](#) bivariate Gamma (chi-square) law.
- Angular and radial parts are independent: $p(\rho_1, u, v) = p_{\text{Fisher}}(\rho_1) p_{\text{Kibble}}(u, v)$.

Remark 22 (Fisher–Kibble decoupling and independence) *By rotational invariance, the empirical correlation and squared norms satisfy*

$$\rho_1 \sim \text{Fisher}(\rho, r), \quad (u, v) \sim \text{Kibble}(r, \rho), \quad \text{and} \quad p(\rho_1, u, v) = p_{\text{Fisher}}(\rho_1) p_{\text{Kibble}}(u, v).$$

Homogeneity and factorization (why angular and radial parts decouple). *For positively 1-homogeneous activations (Assumption 1), the first-layer base kernel separates radial and angular dependence: for all $x, x' \neq 0$,*

$$\Sigma^{(1)}(x, x') = \|x\| \|x'\| \tilde{\Sigma}^{(1)}(\rho), \quad \rho = \frac{\langle x, x' \rangle}{\|x\| \|x'\|},$$

where $\tilde{\Sigma}^{(1)}$ is a scalar function on $[-1, 1]$ (for ReLU it is the standard arc-cosine kernel). Under isotropic Gaussian projections, rotational invariance implies that the empirical correlation $\rho_1 = \cos \angle(x_1, y_1)$ depends only on the angular part of the projected pair, while the norms $(\|x_1\|, \|y_1\|)$ depend only on the radial part; consequently ρ_1 is independent of $(u, v) = (\|x_1\|^2, \|y_1\|^2)$ and the joint law factorizes as stated above. The Fisher distribution density [Fisher \(1915\)](#) is

$$p(\rho_1 | \rho, r) = \frac{(r-2) \Gamma(r-1) (1-\rho^2)^{\frac{r-1}{2}} (1-\rho_1^2)^{\frac{r-4}{2}}}{\sqrt{2\pi} \Gamma(r-\frac{1}{2}) (1-\rho\rho_1)^{r-\frac{3}{2}}} {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; r-\frac{1}{2}; \frac{1+\rho\rho_1}{2}\right), \quad r > 2.$$

For large r , the inverse Gudermann function applied to the complementary angle is approximately normal:

$$\text{gd}^{-1}\left(\frac{\pi}{2} - \arccos(\rho_1)\right) \approx \mathcal{N}\left(\text{arctanh}(\rho), \frac{1}{r-3}\right), \quad \text{hence} \quad \text{Var}(\rho_1) = O(1/r).$$

Kibble’s [Kibble \(1941\)](#) bivariate chi-square law for $(u, v) = (\|x_1\|^2, \|y_1\|^2)$ has density

$$f(u, v) = \frac{(uv)^{\frac{r/2-1}{2}} \exp\left(-\frac{u+v}{2(1-\rho^2)}\right)}{\Gamma(r/2) (2(1-\rho^2))^{\frac{r}{2}+1} \rho^{\frac{r/2-1}{2}}} I_{\frac{r}{2}-1}\left(\frac{\rho\sqrt{uv}}{1-\rho^2}\right), \quad u, v \geq 0,$$

where I_ν is the modified Bessel function of the first kind. The moments are

$$\mathbb{E}[u] = \mathbb{E}[v] = r, \quad \text{Var}(u) = \text{Var}(v) = 2r, \quad \text{Cov}(u, v) = 2r \rho^2.$$

Defining $w_r = \sqrt{uv}/r$, concentration results imply $\mathbb{E}[w_r] \rightarrow 1$ and $\text{Var}(w_r) = O(1/r)$.

Compact three-layer empirical NTK representation. In the three-layer low-rank setting, denote

$$\rho_1 = \cos \angle(\mathbf{h}^{(1)}(x), \mathbf{h}^{(1)}(x')), \quad w_r = \frac{\|\mathbf{h}^{(1)}(x)\| \|\mathbf{h}^{(1)}(x')\|}{r}.$$

Then the empirical three-layer NTK admits the compact form

$$\Theta^{(2)}(x, x') = 1 + \frac{1}{r} \Theta^{(1)}(\rho_1) \left(1 - \frac{\arccos(\rho_1)}{\pi}\right) + \frac{1}{r} w_r \Sigma^{(1)}(\rho_1),$$

with $\Sigma^{(1)}(u) = \frac{1}{\pi} (\sqrt{1-u^2} + u(1 - \arccos u))$.

Proof Let $x, y \in \mathbb{R}^d$ be fixed input vectors with correlation $\rho = \langle x, y \rangle / (\|x\| \|y\|)$. Consider a rank- r random projection matrix $P \in \mathbb{R}^{r \times d}$ with entries $P_{ij} \sim \mathcal{N}(0, 1/d)$ i.i.d., so that $x_1 = Px$ and $y_1 = Py$ are the projected vectors.

Since P has i.i.d. Gaussian entries, the projected vectors $x_1, y_1 \in \mathbb{R}^r$ form a ρ -correlated bivariate Gaussian pair. Specifically, for each component $i = 1, \dots, r$, the pairs (x_{1i}, y_{1i}) are i.i.d. with joint distribution

$$\begin{pmatrix} x_{1i} \\ y_{1i} \end{pmatrix} \sim \mathcal{N}\left(\mathbf{0}, \begin{pmatrix} \|x\|^2/d & \rho \|x\| \|y\|/d \\ \rho \|x\| \|y\|/d & \|y\|^2/d \end{pmatrix}\right). \quad (29)$$

By rotational invariance of the Gaussian projection and the fact that the correlation ρ is preserved under orthogonal transformations, we can assume without loss of generality that the covariance structure factors into angular and radial components.

Angular part (Fisher distribution). The empirical correlation is defined as

$$\rho_1 = \frac{\langle x_1, y_1 \rangle}{\|x_1\| \|y_1\|} = \frac{\sum_{i=1}^r x_{1i} y_{1i}}{\sqrt{\sum_{i=1}^r x_{1i}^2} \sqrt{\sum_{i=1}^r y_{1i}^2}}. \quad (30)$$

This is the sample correlation coefficient of r pairs of correlated Gaussian random variables. Fisher [Fisher \(1915\)](#); [Hotelling \(1953\)](#) showed that the distribution of ρ_1 depends only on the true correlation ρ and the sample size r , with density given in Remark 22. The key observation is that ρ_1 is a function purely of the angles between the projected vectors, independent of their magnitudes.

Radial part (Kibble distribution). The squared norms $u = \|x_1\|^2 = \sum_{i=1}^r x_{1i}^2$ and $v = \|y_1\|^2 = \sum_{i=1}^r y_{1i}^2$ are sums of correlated chi-square variables. Since (x_{1i}, y_{1i}) are ρ -correlated Gaussians, the bivariate distribution of (u, v) follows Kibble's [Kibble \(1941\)](#) bivariate chi-square law. The joint density involves the modified Bessel function $I_\nu(z)$, which appears because the dot product $\langle x_1, y_1 \rangle = \sum_{i=1}^r x_{1i} y_{1i}$ can be expressed as a weighted sum of products of correlated normals, whose distribution relates to Bessel functions through the generating function of the bivariate chi-square distribution.

Specifically, the joint characteristic function of (u, v) is

$$\phi(s, t) = \mathbb{E}[e^{i(su+tv)}] = (1 - 2i(1 - \rho^2)(s + t) - 4\rho^2 st)^{-r/2}, \quad (31)$$

and the inverse Fourier transform yields Kibble's density, which contains the modified Bessel function $I_{\frac{r}{2}-1}$ as stated in Remark 22. The Bessel function arises from the integral representation of the bivariate chi-square density when the correlation $\rho \neq 0$.

Independence of angular and radial parts. Rotational invariance of isotropic Gaussian projections implies that $\rho_1 = \cos \angle(x_1, y_1)$ and the norms $(\|x_1\|, \|y_1\|)$ are independent, yielding the factorization $p(\rho_1, u, v) = p_{\text{Fisher}}(\rho_1) \cdot p_{\text{Kibble}}(u, v)$. This is the classical independence of the sample correlation from sample variances in bivariate normal data. ■

C.3. Open directions: deep mean NTK recursion and effective depth

A full treatment of the deep ($L \rightarrow \infty$) mean NTK recursion requires: (i) a rigorous recursion for $\tilde{\Theta}^{(\ell)}$ with explicit covariance bounds; (ii) verification that $\tilde{\Sigma}^{(\ell)}$ remains uniformly bounded in ℓ and ρ_ℓ ; (iii) extension of Fisher–Kibble decoupling to nested layers to control mixed terms such as $\mathbb{E}[\Theta^{(\ell-1)} \dot{\Sigma}^{(\ell)}]$ (where dependence through the deep GP chain prevents naive factorization). The exponential suppression of early-layer contributions suggests that very deep MMNNs may exhibit an “effective depth” phenomenon similar to ResNets, where only the top layers meaningfully contribute to the NTK.

Lemma 23 (Nonasymptotic concentration of Gaussian sample cosine) *Let $r \geq 1$ and let $(X, Y) \in \mathbb{R}^r \times \mathbb{R}^r$ have i.i.d. coordinates $\{(X_i, Y_i)\}_{i=1}^r$, each distributed as a centered bivariate normal with $\mathbb{E}[X_i^2] = \mathbb{E}[Y_i^2] = 1$ and $\mathbb{E}[X_i Y_i] = \rho \in (-1, 1)$. Define the empirical cosine (sample correlation without centering)*

$$\hat{\rho}_r = \frac{\langle X, Y \rangle}{\|X\| \|Y\|}.$$

Then there exist absolute constants $c, C > 0$ such that for all $t \in (0, 1)$,

$$\mathbb{P}(|\hat{\rho}_r - \rho| \geq t) \leq 6 \exp(-c r t^2), \quad \mathbb{E}[(\hat{\rho}_r - \rho)^2] \leq \frac{C}{r}.$$

Proof Write

$$S_{xx} = \frac{1}{r} \|X\|^2, \quad S_{yy} = \frac{1}{r} \|Y\|^2, \quad S_{xy} = \frac{1}{r} \langle X, Y \rangle, \quad \text{so that} \quad \hat{\rho}_r = \frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}}.$$

Each of $S_{xx} - 1$, $S_{yy} - 1$, and $S_{xy} - \rho$ is an average of i.i.d. centered sub-exponential random variables (since $X_i^2 - 1$, $Y_i^2 - 1$, and $X_i Y_i - \rho$ have finite ψ_1 norms for Gaussian data). Therefore, by Bernstein’s inequality, there exist absolute constants $c_0 > 0$ and $C_0 < \infty$ such that for all $t \in (0, 1)$,

$$\mathbb{P}(|S_{xx} - 1| \geq t) \leq 2e^{-c_0 r t^2}, \quad \mathbb{P}(|S_{yy} - 1| \geq t) \leq 2e^{-c_0 r t^2}, \quad \mathbb{P}(|S_{xy} - \rho| \geq t) \leq 2e^{-c_0 r t^2}.$$

On the event $E_t = \{|S_{xx} - 1| \leq t, |S_{yy} - 1| \leq t\}$ with $t \leq 1/2$, we have $S_{xx}, S_{yy} \in [1/2, 3/2]$, hence

$$\left| \frac{1}{\sqrt{S_{xx} S_{yy}}} - 1 \right| \leq C_0 (|S_{xx} - 1| + |S_{yy} - 1|) \leq 2C_0 t$$

for an absolute constant C_0 . Using

$$\hat{\rho}_r - \rho = \frac{S_{xy} - \rho}{\sqrt{S_{xx} S_{yy}}} + \rho \left(\frac{1}{\sqrt{S_{xx} S_{yy}}} - 1 \right),$$

we obtain on E_t the bound $|\hat{\rho}_r - \rho| \leq C_1(|S_{xy} - \rho| + t)$ for an absolute constant C_1 . Taking $t \asymp t'$ and applying a union bound over the three Bernstein tails yields $\mathbb{P}(|\hat{\rho}_r - \rho| \geq t') \leq 6e^{-crt'^2}$ for $t' \in (0, 1)$ and absolute $c > 0$. The second-moment bound follows by integrating the tail bound: $\mathbb{E}[(\hat{\rho}_r - \rho)^2] = \int_0^\infty 2u \mathbb{P}(|\hat{\rho}_r - \rho| \geq u) du \leq C/r$. $\blacksquare$

C.4. Rank-driven concentration

Theorem 24 (Concentration in rank for homogeneous activation) *Under the homogeneous activation assumption (Assumption 1), let $x_1, y_1 \in \mathbb{R}^r$ be the rank- r Gaussian projections of inputs x, y , with arbitrary fixed correlation $\rho \in [-1, 1]$. Define*

$$W = \frac{\|x_1\| \|y_1\|}{r}.$$

Then for any $\epsilon \in (0, 1)$,

$$\mathbb{P}(|W - 1| \geq \epsilon) \leq 4 \exp\left(-\frac{r\epsilon^2}{8}\right).$$

Moreover, for larger deviations $\epsilon \geq 1$, there exist absolute constants $c_1, c_2 > 0$ such that

$$\mathbb{P}(|W - 1| \geq \epsilon) \leq c_1 \exp(-c_2 r \epsilon).$$

In particular, the radial component of the RF-LR NTK concentrates exponentially fast in r , yielding high-dimensional control of RKHS fluctuations.

Proof Let $x_1, y_1 \in \mathbb{R}^r$ be Gaussian projections with arbitrary fixed correlation $\rho \in [-1, 1]$. Define $U = \|x_1\|/\sqrt{r}$, $V = \|y_1\|/\sqrt{r}$, and $W = UV = \|x_1\| \|y_1\|/r$. We prove that for $\epsilon \in (0, 1)$,

$$\mathbb{P}(|W - 1| \geq \epsilon) \leq 4 \exp\left(-\frac{r\epsilon^2}{8}\right), \quad (32)$$

and that for $\epsilon \geq 1$ there exist constants $c_1, c_2 > 0$ with $\mathbb{P}(|W - 1| \geq \epsilon) \leq c_1 e^{-c_2 r \epsilon}$.

The Euclidean norm is 1-Lipschitz, so by Gaussian concentration, for all $\delta > 0$,

$$\mathbb{P}(|U - 1| \geq \delta) \leq 2 \exp\left(-\frac{r\delta^2}{2}\right), \quad \mathbb{P}(|V - 1| \geq \delta) \leq 2 \exp\left(-\frac{r\delta^2}{2}\right). \quad (33)$$

We use

$$|UV - 1| \leq |U - 1| + |V - 1| + |U - 1| |V - 1|. \quad (34)$$

Fix $\epsilon \in (0, 1)$ and set $\delta = \epsilon/2$. On the event $\{|U - 1| < \delta, |V - 1| < \delta\}$, we have

$$|UV - 1| \leq 2\delta + \delta^2 \leq \epsilon. \quad (35)$$

Therefore,

$$\mathbb{P}(|UV - 1| \geq \epsilon) \leq \mathbb{P}(|U - 1| \geq \delta) + \mathbb{P}(|V - 1| \geq \delta) \leq 4 \exp\left(-\frac{r\epsilon^2}{8}\right). \quad (36)$$

$\blacksquare$

Corollary 25 (Concentration bound for three-layer NTK) *Under Assumption 1, let $\hat{\Theta}_r^{(2)}(x, x') = \Psi(\hat{\rho}_r, \hat{w}_r)$ be the empirical three-layer NTK, where $\hat{\rho}_r$ is the sample correlation and $\hat{w}_r = \|x_1\| \|y_1\|/r$, with $x_1, y_1 \in \mathbb{R}^r$ being the rank- r Gaussian projections of inputs x, y with population correlation $\rho \in [-1, 1]$. Let $K_\infty(\rho) = \Psi(\rho, 1)$ be the deterministic kernel limit. Then for any $\epsilon \in (0, 1)$, there exist constants $C_1, C_2 > 0$ depending on ρ and the kernel smoothness such that*

$$\mathbb{P}\left(\left|\hat{\Theta}_r^{(2)}(x, x') - K_\infty(\rho)\right| \geq \epsilon\right) \leq C_1 \exp(-C_2 r \epsilon^2).$$

C.5. Proof of Corollary 25: Concentration bound for three-layer NTK

Proof We prove that the empirical three-layer NTK $\hat{\Theta}_r^{(2)}(x, x') = \Psi(\hat{\rho}_r, \hat{w}_r)$ concentrates exponentially fast around its deterministic limit $K_\infty(\rho) = \Psi(\rho, 1)$ as $r \rightarrow \infty$.

Recall that the kernel function is defined as:

$$\Psi(u, w) = \Theta^{(1)}(u) \left(1 - \frac{\arccos(u)}{\pi}\right) + w \Sigma^{(1)}(u) + 1, \quad (37)$$

where $\hat{\rho}_r$ is the sample correlation and $\hat{w}_r = \|x_1\| \|y_1\|/r$. By Lemma 23, $\hat{\rho}_r$ concentrates around ρ with sub-Gaussian tails at rate $r^{-1/2}$, and by Theorem 24, $\hat{w}_r$ concentrates around 1 with sub-Gaussian tails.

The function Ψ is differentiable in both arguments. For fixed $\rho \in (-1, 1)$, we have:

$$\partial_u \Psi(\rho, 1) = \partial_\rho K_\infty(\rho) = O(1), \quad (38)$$

$$\partial_w \Psi(\rho, 1) = \Sigma^{(1)}(\rho) = O(1). \quad (39)$$

Since $\Theta^{(1)}$, $\Sigma^{(1)}$, and $\arccos$ are smooth on $(-1, 1)$, there exist Lipschitz constants $L_u, L_w > 0$ (depending on ρ) such that for $u, u' \in [\rho - \delta, \rho + \delta]$ and $w, w' \in [1 - \delta, 1 + \delta]$ with $\delta > 0$ small:

$$|\Psi(u, w) - \Psi(u', w')| \leq L_u |u - u'| + L_w |w - w'|. \quad (40)$$

We combine radial and angular concentration by using a union bound and the Lipschitz property:

$$|\hat{\Theta}_r^{(2)}(x, x') - K_\infty(\rho)| = |\Psi(\hat{\rho}_r, \hat{w}_r) - \Psi(\rho, 1)| \leq L_u |\hat{\rho}_r - \rho| + L_w |\hat{w}_r - 1|. \quad (41)$$

Therefore, for $\epsilon \in (0, 1)$:

$$\mathbb{P}\left(|\hat{\Theta}_r^{(2)}(x, x') - K_\infty(\rho)| \geq \epsilon\right) \leq \mathbb{P}\left(|\hat{\rho}_r - \rho| \geq \frac{\epsilon}{2L_u}\right) + \mathbb{P}\left(|\hat{w}_r - 1| \geq \frac{\epsilon}{2L_w}\right). \quad (42)$$

Now we prove the sub-Gaussian concentration of the sample correlation via Hanson–Wright.

Lemma 26 (Sub-Gaussian concentration of the sample correlation via Hanson–Wright) *Let $(X_i, Y_i)_{i=1}^r$ be i.i.d. centered Gaussian pairs with unit variances and correlation $\rho \in (-1, 1)$. Define*

$$S_{xx} = \frac{1}{r} \sum_{i=1}^r X_i^2, \quad S_{yy} = \frac{1}{r} \sum_{i=1}^r Y_i^2, \quad S_{xy} = \frac{1}{r} \sum_{i=1}^r X_i Y_i, \quad \hat{\rho}_r = \frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}}. \quad (43)$$

There exist absolute constants $c, C > 0$ such that for all $t \in (0, 1)$,

$$\mathbb{P}(|\hat{\rho}_r - \rho| \geq t) \leq C \exp(-c r t^2). \quad (44)$$

Proof

Write $Y = \rho X + \sqrt{1 - \rho^2} Z$ with $X = (X_i)_{i \leq r}$ and $Z = (Z_i)_{i \leq r}$ independent, i.i.d. $\mathcal{N}(0, 1)$. Then

$$S_{xy} - \rho = \rho(S_{xx} - 1) + \underbrace{\sqrt{1 - \rho^2} \frac{1}{r} \sum_{i=1}^r X_i Z_i}_{=: T_r}. \quad (45)$$

From [Vershynin \(2018\)](#) Theorem 6.2.2 page 183, the bilinear Hanson–Wright inequality (for independent sub-Gaussian X, Z and fixed $M \in \mathbb{R}^{r \times r}$) states

$$\mathbb{P}\left(|X^\top M Z| \geq u\right) \leq 2 \exp\left[-c \min\left(\frac{u^2}{K^4 \|M\|_F^2}, \frac{u}{K^2 \|M\|}\right)\right], \quad (46)$$

where $K = O(1)$ is the Orlicz sub-gaussian norm of the coordinates. For $M = I_r$ we have $\|M\|_F^2 = r$ and $\|M\| = 1$, hence with $u = rt$ and $t \in (0, 1)$,

$$\mathbb{P}\left(\left|\frac{1}{r} X^\top Z\right| \geq t\right) \leq 2 \exp(-c r t^2). \quad (47)$$

Thus T_r is sub-Gaussian at rate $\exp(-c r t^2)$:

$$\mathbb{P}(|T_r| \geq t) \leq 2 \exp(-c r t^2). \quad (48)$$

For $S_{xx} = (1/r)\|X\|^2$ and $S_{yy} = (1/r)\|Y\|^2$, Laurent–Massart Gaussian quadratic chaos inequality in (4.1) page 1325 [Laurent and Massart \(2000\)](#) yields, for any $s \in (0, 1)$,

$$\mathbb{P}(|S_{xx} - 1| \geq s) \leq 2 \exp(-c r s^2), \quad \mathbb{P}(|S_{yy} - 1| \geq s) \leq 2 \exp(-c r s^2). \quad (49)$$

Let $f(a, b, c) = a/\sqrt{bc}$. A first-order expansion at $(\rho, 1, 1)$ yields

$$|\hat{\rho}_r - \rho| = |f(S_{xy}, S_{xx}, S_{yy}) - f(\rho, 1, 1)| \leq |S_{xy} - \rho| + \frac{|\rho|}{2} (|S_{xx} - 1| + |S_{yy} - 1|) + R, \quad (50)$$

where the remainder $R = O((|S_{xx} - 1| + |S_{yy} - 1|)^2)$ is negligible on the event $\{|S_{xx} - 1| \vee |S_{yy} - 1| \leq c_0\}$ (for some absolute c_0 , e.g., $1/4$). Choosing $s = \kappa t$ for a sufficiently small absolute $\kappa > 0$, and splitting $|S_{xy} - \rho|$ as in step 1, a union bound gives

$$\mathbb{P}(|\hat{\rho}_r - \rho| \geq t) \leq \underbrace{\mathbb{P}(|T_r| \geq c_1 t)}_{\leq 2e^{-c r t^2}} + \underbrace{\mathbb{P}(|S_{xx} - 1| \geq c_2 t)}_{\leq 2e^{-c r t^2}} + \underbrace{\mathbb{P}(|S_{yy} - 1| \geq c_2 t)}_{\leq 2e^{-c r t^2}} + \mathbb{P}(R \not\leq t). \quad (51)$$

The last term is absorbed by adjusting κ (small-probability event controlled by the same inequalities). Therefore, for absolute constants $c, C > 0$,

$$\mathbb{P}(|\hat{\rho}_r - \rho| \geq t) \leq C \exp(-c r t^2), \quad t \in (0, 1). \quad (52)$$

■

Remark 27 (“Bessel” variant: sum of Gaussian products) Each product $X_i Z_i$ has a symmetric sub-exponential tail (density involving the Bessel function K_0). By Bernstein’s inequality for i.i.d. sub-exponential variables, $\frac{1}{r} \sum_{i=1}^r X_i Z_i$ satisfies the same $\exp(-crt^2)$ bound for $t \in (0, 1)$, yielding an alternative proof of the bilinear step.

By Theorem 24, for $\epsilon/(2L_w) \in (0, 1)$:

$$\mathbb{P}\left(|\hat{w}_r - 1| \geq \frac{\epsilon}{2L_w}\right) \leq 4 \exp\left(-\frac{r\epsilon^2}{32L_w^2}\right). \quad (53)$$

Combining both terms:

$$\begin{aligned} \mathbb{P}\left(|\hat{\Theta}_r^{(2)}(x, x') - K_\infty(\rho)| \geq \epsilon\right) &\leq \underbrace{\mathbb{P}\left(|\hat{\rho}_r - \rho| \geq \frac{\epsilon}{2L_u}\right)}_{\leq C \exp\left(-\frac{cr\epsilon^2}{4L_u^2}\right)} + \underbrace{\mathbb{P}\left(|\hat{w}_r - 1| \geq \frac{\epsilon}{2L_w}\right)}_{\leq 4 \exp\left(-\frac{r\epsilon^2}{32L_w^2}\right)} \end{aligned} \quad (54)$$

so that

$$\mathbb{P}\left(|\hat{\Theta}_r^{(2)}(x, x') - K_\infty(\rho)| \geq \epsilon\right) \leq C \exp\left(-\frac{cr\epsilon^2}{4L_u^2}\right) + 4 \exp\left(-\frac{r\epsilon^2}{32L_w^2}\right). \quad (55)$$

Taking $C_1 = C + 4$ and $C_2 = \min(c/(4L_u^2), 1/(32L_w^2))$, we obtain

$$\mathbb{P}\left(|\hat{\Theta}_r^{(2)}(x, x') - K_\infty(\rho)| \geq \epsilon\right) \leq C_1 \exp(-C_2 r \epsilon^2). \quad (56)$$

■

C.6. Proof of Corollary 20: Mean NTK over Fisher–Kibble has same RKHS

Proof We prove that under Assumption 1, the mean NTK $\tilde{\Theta}^{(2)}$ for the three-layer case (obtained by taking expectations over Fisher and Kibble distributions) is a zonal kernel that induces the same RKHS as the shallow ReLU kernel.

Puiseux expansion of mean NTK near $\rho = 1$: The key observation is that the mean NTK $\tilde{\Theta}^{(2)}(\rho)$ is a zonal kernel (depends only on ρ). Near $\rho = 1$ (for $t \geq 0$), we need to compute the Puiseux expansion of $\mathbb{E}[\arccos(\hat{\rho}_r)]$ when $\rho = 1 - t$, where $\hat{\rho}_r \sim \text{Fisher}(1 - t, r)$.

This is highly non-trivial because we must compute:

$$\mathbb{E}[\arccos(\hat{\rho}_r)] = \int_{-1}^1 \arccos(u) p_{\text{Fisher}}(u \mid \rho = 1 - t, r) du, \quad (57)$$

where the full Fisher distribution density is:

$$p_{\text{Fisher}}(u \mid \rho, r) = \frac{(r-2) \Gamma(r-1) (1-\rho^2)^{\frac{r-1}{2}} (1-u^2)^{\frac{r-4}{2}}}{\sqrt{2\pi} \Gamma(r-\frac{1}{2}) (1-\rho u)^{r-\frac{3}{2}}} {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; r-\frac{1}{2}; \frac{1+\rho u}{2}\right), \quad (58)$$

for $r > 2$, with ${}_2F_1$ the hypergeometric function.

Making the change of variables $v = 1 - u$ and $s = 1 - t$, so $\rho = s = 1 - t$ and $u = 1 - v$, we have:

$$\mathbb{E}[\arccos(\hat{\rho}_r)] = \int_0^2 \arccos(1 - v) p_{\text{Fisher}}(1 - v \mid s, r) dv. \quad (59)$$

Near $v = 0$ and $t = 0$, we analyze the asymptotic behavior of the Fisher density. For $\rho = s = 1 - t$ and $u = 1 - v$ with $t, v \rightarrow 0^+$:

$$1 - \rho^2 = 1 - (1 - t)^2 = 2t - t^2 = 2t(1 - t/2), \quad (60)$$

$$1 - u^2 = 1 - (1 - v)^2 = 2v - v^2 = 2v(1 - v/2), \quad (61)$$

$$1 - \rho u = 1 - (1 - t)(1 - v) = t + v - tv = t + v + O(tv). \quad (62)$$

The Fisher density near $v = 0$ and $t = 0$ behaves as:

$$p_{\text{Fisher}}(1 - v \mid 1 - t, r) \sim \frac{(r - 2) \Gamma(r - 1) (2t)^{\frac{r-1}{2}} (2v)^{\frac{r-4}{2}}}{\sqrt{2\pi} \Gamma(r - \frac{1}{2}) (t + v)^{r-\frac{3}{2}}} {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; r - \frac{1}{2}; 1 - \frac{t+v}{2}\right). \quad (63)$$

For the hypergeometric function near its argument $1 - (t + v)/2 \rightarrow 1^-$, we invoke the connection formula in DLMF §15.8.2 [Olver et al. \(2024\)](#) together with the fact that $c - a - b = r - \frac{3}{2} > 0$ for $r \geq 2$ and is a half-integer (hence no logarithmic term appears). Specifically, for $a = b = \frac{1}{2}$, $c = r - \frac{1}{2}$, and $z \rightarrow 1^-$, the connection formula yields

$${}_2F_1(a, b; c; z) = \frac{\Gamma(c) \Gamma(c - a - b)}{\Gamma(c - a) \Gamma(c - b)} + O((1 - z)^{c-a-b}). \quad (64)$$

Setting $z = 1 - \frac{t+v}{2}$ we obtain, uniformly for $t, v \rightarrow 0^+$,

$${}_2F_1\left(\frac{1}{2}, \frac{1}{2}; r - \frac{1}{2}; 1 - \frac{t+v}{2}\right) = C_1(r) + O((t + v)^{r-\frac{3}{2}}), \quad (65)$$

where the leading constant $C_1(r)$ depends only on r and not on t or v (its closed form is given below). Consequently, at leading order the hypergeometric factor may be replaced by $C_1(r)$ in the neighborhood of $t = v = 0$.

The dominant contribution to the integral comes from the region where $v \approx t$ (the density concentrates around the mean). Near this region, $\arccos(1 - v) = \sqrt{2v} + O(v^{3/2})$, so:

$$\mathbb{E}[\arccos(\hat{\rho}_r)] = \int_0^\infty \sqrt{2v} \cdot \frac{(r - 2) \Gamma(r - 1) (2t)^{\frac{r-1}{2}} (2v)^{\frac{r-4}{2}}}{\sqrt{2\pi} \Gamma(r - \frac{1}{2}) (t + v)^{r-\frac{3}{2}}} C_1(r) dv + O(t^{3/2}). \quad (66)$$

Making the substitution $w = v/t$, we have $v = tw$, $dv = t dw$, and $t + v = t(1 + w)$:

$$\mathbb{E}[\arccos(\hat{\rho}_r)] = \int_0^\infty \sqrt{2tw} \cdot \frac{(r - 2) \Gamma(r - 1) (2t)^{\frac{r-1}{2}} (2tw)^{\frac{r-4}{2}}}{\sqrt{2\pi} \Gamma(r - \frac{1}{2}) (t(1 + w))^{r-\frac{3}{2}}} C_1(r) \cdot t dw + O(t^{3/2}). \quad (67)$$

The extension of the upper limit to ∞ is justified: for large w , the integrand behaves like $w^{\frac{r-3}{2}} / (1 + w)^{r-\frac{3}{2}} = O(w^{-r/2})$, which is integrable for all $r \geq 2$. Hence the tail beyond any

fixed cutoff W contributes $o(t^{1/2})$ uniformly as $t \rightarrow 0^+$, and is absorbed into the $O(t^{3/2})$ remainder.

Simplifying the powers of t :

$$= \sqrt{2t} \cdot t^{\frac{r-1}{2} + \frac{r-4}{2} + 1 - (r - \frac{3}{2})} \cdot \frac{(r-2) \Gamma(r-1) (2)^{\frac{r-1}{2}} (2w)^{\frac{r-4}{2}}}{\sqrt{2\pi} \Gamma(r - \frac{1}{2}) (1+w)^{r-\frac{3}{2}}} C_1(r) \cdot \sqrt{w} dw + O(t^{3/2}), \quad (68)$$

where the exponent is $\frac{r-1}{2} + \frac{r-4}{2} + 1 - (r - \frac{3}{2}) = 0$, so:

$$\mathbb{E}[\arccos(\hat{\rho}_r)] = \sqrt{2t} \cdot I(r) + O(t^{3/2}), \quad (69)$$

where

$$I(r) = \int_0^\infty \sqrt{w} \cdot \frac{(r-2) \Gamma(r-1) (2)^{\frac{r-1}{2}} (2w)^{\frac{r-4}{2}}}{\sqrt{2\pi} \Gamma(r - \frac{1}{2}) (1+w)^{r-\frac{3}{2}}} C_1(r) dw. \quad (70)$$

Evaluating the integral yields the exact constant

$$I(r) = \frac{(r-2) 2^{r-\frac{5}{2}} \Gamma(\frac{r-1}{2}) \Gamma(\frac{r}{2} - 1)}{\sqrt{2\pi} \Gamma(r-1)} C_1(r), \quad C_1(r) = \frac{\Gamma(r - \frac{1}{2}) \Gamma(r - \frac{3}{2})}{\Gamma(r-1)^2}. \quad (71)$$

Therefore,

$$\mathbb{E}[\arccos(\hat{\rho}_r)] = \sqrt{2t} I(r) + O(t^{3/2}). \quad (72)$$

Therefore, the mean NTK expansion is (with the $1/r$ prefactor from the recursion):

$$\tilde{\Theta}^{(2)}(1-t) = 1 + \frac{1}{r} (K_\infty(1) - 1) - \frac{1}{r} \left[\frac{2}{\pi} + \frac{\sqrt{2}}{2\pi} I(r) \right] t^{1/2} + O(t^{3/2}). \quad (73)$$

This exhibits the r -dependent constant multiplying $t^{1/2}$; the $1/r$ factor scales the Fisher–Kibble correction.

Remark 28 (Closed form via Beta function and asymptotics) *Using the change of variables $w = v/t$ and Euler’s Beta integral, the inner integral appearing in $I(r)$ can be evaluated explicitly:*

$$\int_0^\infty \frac{w^{\frac{r-3}{2}}}{(1+w)^{r-\frac{3}{2}}} dw = B\left(\frac{r-1}{2}, \frac{r}{2} - 1\right) = \frac{\Gamma(\frac{r-1}{2}) \Gamma(\frac{r}{2} - 1)}{\Gamma(r - \frac{3}{2})}. \quad (74)$$

Substituting this identity into the pre-factors gives the stated closed form for $I(r)$ in (75) below, which matches the expression above:

$$I(r) = \frac{(r-2) 2^{r-\frac{5}{2}}}{\sqrt{2\pi}} \frac{\Gamma(\frac{r-1}{2}) \Gamma(\frac{r}{2} - 1)}{\Gamma(r-1)} C_1(r), \quad C_1(r) = \frac{\Gamma(r - \frac{1}{2}) \Gamma(r - \frac{3}{2})}{\Gamma(r-1)^2}. \quad (75)$$

The asymptotic scaling $I(r) \sim C/\sqrt{r}$ as $r \rightarrow \infty$ is established in Proposition 29 (Appendix C.7). In particular, the leading Puiseux coefficient

$$1 - \left[\frac{2}{\pi} + \frac{\sqrt{2}}{2\pi} I(r) \right] t^{1/2} + O(t^{3/2}) \quad (76)$$

is strictly positive and depends smoothly on r . This confirms that the endpoint exponent $\frac{1}{2}$ (hence the RKHS spectral decay rate) is unchanged by the Fisher–Kibble expectation, while the multiplicative constant is an $O(1)$ perturbation of the shallow ReLU constant $2/\pi$.

RKHS equivalence via endpoint behavior. The mean NTK $\tilde{\Theta}^{(2)}(\rho)$ is a zonal kernel with a Puiseux expansion that differs from the deterministic kernel $K_\infty(\rho)$ due to the Fisher–Kibble expectations. However, the leading-order behavior near $\rho = 1$ is preserved: both kernels have the same $t^{1/2}$ leading term in their Puiseux expansions.

By Theorem 1 of [Bietti and Bach \(2021\)](#), the RKHS spectral decay is determined by the leading-order Puiseux expansion coefficient (the $t^{1/2}$ term). Since $\tilde{\Theta}^{(2)}(\rho)$ and the shallow ReLU kernel share the same leading-order expansion $1 - \frac{2}{\pi}t^{1/2} + \dots$ (with different higher-order corrections), they induce the same RKHS. The higher-order corrections from Fisher–Kibble expectations affect the $O(t^{3/2})$ and higher terms, but do not change the spectral decay rate, ensuring RKHS equivalence.

Extension to general L : open problem. The extension to general depth $L \geq 3$ remains open. A rigorous proof would require: (i) the conditional law of the Fisher chain $(\rho, \rho_1, \dots, \rho_{L-2})$ given $\rho_{L-1} = u$, or equivalently the $(L-1)$ -fold marginal distribution of ρ_{L-1} given ρ ; (ii) an explicit Laplace-type expansion of the integral $\int_{-1}^1 \dot{\Sigma}^{(L)}(u) \mathbb{E}[\Theta^{(L-1)} \mid \rho_{L-1} = u] p_{\text{Fisher}}(u \mid \rho, r) du$ (and analogous terms) as $\rho = 1 - t \rightarrow 1^-$. The conditional mean $\mathbb{E}[\Theta^{(L-1)} \mid \rho_{L-1} = u]$ is not equal to $\tilde{\Theta}^{(L-1)}(u)$, since the former conditions on the chain endpoint while the latter fixes the input; verifying the Puiseux structure of the conditional expectation would require a detailed analysis of the backward Fisher-chain density. We conjecture that the same $t^{1/2}$ leading exponent holds for all L , and that the mean NTK induces the same RKHS; we leave a complete proof for future work. ■

C.7. Asymptotic scaling of $I(r)$

The Fisher–Kibble integral $I(r)$ defined in (75) governs the r -dependent coefficient in the mean NTK’s Puiseux expansion near $\rho = 1$. The following proposition gives its scaling as $r \rightarrow \infty$.

Proposition 29 ($I(r)$ decays as $1/\sqrt{r}$) *Let $I(r)$ be as in (75) with $C_1(r) = \Gamma(r - \frac{1}{2})\Gamma(r - \frac{3}{2})/\Gamma(r - 1)^2$. Then*

$$I(r) \sim \frac{C}{\sqrt{r}} \quad \text{as } r \rightarrow \infty, \quad \text{for some } C > 0, \quad (77)$$

i.e., $I(r) = O(1/\sqrt{r})$. For fixed $r \geq 3$, $I(r)$ is bounded and strictly positive.

Proof Using the Beta representation $B((r-1)/2, r/2-1) = \Gamma((r-1)/2)\Gamma(r/2-1)/\Gamma(r-3/2)$, the closed form reduces to $I(r) = (r-2)2^{r-5/2}/\sqrt{2\pi} \cdot B \cdot \Gamma(r-3/2)\Gamma(r-1/2)/\Gamma(r-1)^3$. By Stirling’s formula and the asymptotics $B(a, b) \sim \sqrt{2\pi} a^{a-1/2}b^{b-1/2}/(a+b)^{a+b-1/2}$ when $a, b \rightarrow \infty$ with $a \sim b \sim r/2$, one has $B((r-1)/2, r/2-1) \sim \sqrt{2\pi} 2^{-r}/\sqrt{r}$. The Gamma-ratio $\Gamma(r-1/2)\Gamma(r-3/2)/\Gamma(r-1)^2 \sim 1$. Combining these yields $I(r) \sim C/\sqrt{r}$. ■

Implications: (i) the Puiseux coefficient $c_1(r) = -[2/\pi + (\sqrt{2}/(2\pi))I(r)]$ satisfies $c_1(r) \rightarrow -2/\pi$ as $r \rightarrow \infty$, recovering the shallow ReLU constant; (ii) the Fisher–Kibble correction to the mean NTK is $O(1/(r\sqrt{r}))$ in the $t^{1/2}$ term when $I(r) \sim C/\sqrt{r}$.

Appendix D. Proofs for depth scaling

D.1. Correlation propagation and inverse cosine distances

For ReLU activation, the RF-LR architecture uses the same EOC parameterization as standard MLPs Hayou et al. (2018). The forward correlation map induced by ReLU is the same as in Terjék and González-Sánchez (2025); in particular, if we define a deterministic cosine recursion $\rho_k = \varrho(\rho_{k-1})$ (the infinite-width/full-rank idealization), then for ReLU $\Delta_\phi = 1/2$ (i.e. $a = b = 1/\sqrt{2}$ in (a, b) -ReLU notation).

Definition 30 (Cosine map for ReLU at EOC) For ReLU, set $\Delta_\phi = 1/2$. The cosine map $\varrho : [-1, 1] \rightarrow [-1, 1]$ and its derivative ϱ' satisfy

$$\varrho(\rho) = \rho + \Delta_\phi \frac{2}{\pi} (\sqrt{1 - \rho^2} - \rho \arccos(\rho)), \quad \varrho'(\rho) = 1 - \Delta_\phi \frac{2}{\pi} \arccos(\rho).$$

We write $\rho_k = \varrho(\rho_{k-1})$ for the (deterministic) limiting correlation at layer k .

To a cosine $\rho \in (-1, 1)$, we associate the squared cosine distance $z = (1 - \rho)/2 \in (0, 1)$ and the inverse cosine distance $w = z^{-1/2} \in (1, \infty)$.

Definition 31 (Squared and inverse cosine distance maps) Define $\zeta : [0, 1] \rightarrow [0, 1]$ by $\zeta(z) = (1 - \varrho(1 - 2z))/2$ and $\omega : (1, \infty) \rightarrow (1, \infty)$ by $\omega(w) = \zeta(w^{-2})^{-1/2}$. Then ω is convex and satisfies

$$\omega(w) = w + \Delta_\phi \frac{4}{3\pi} + \frac{3}{2} \left(\Delta_\phi \frac{4}{3\pi} \right)^2 w^{-1} + O(w^{-2}).$$

Proposition 32 (Inverse cosine distance propagation for RF-LR) Given $w \in (1, \infty)$, for $k \in \mathbb{N}$ we have

$$|\omega^{ok}(w) - (w + \Delta_\phi \frac{4}{3\pi}(k-1) + \Delta_\phi \frac{2}{\pi} \log(\Delta_\phi^{-1} \frac{3\pi}{4} w + k - 1))| \leq O(1).$$

Proof This is a statement about the iterates of the inverse cosine distance map ω , so what we must justify is that ω is indeed the correct map governing the RF-LR correlation recursion.

Step 1: RF-LR correlation map (idealized) equals the ReLU cosine map. Fix a layer $\ell \geq 2$ and two inputs (x, x') . Write the RF-LR layer as

$$\mathbf{h}^{(\ell)}(x) = \frac{1}{\sqrt{n_\ell}} \sum_{j=1}^{n_\ell} A_j^{(\ell)} \boldsymbol{\sigma}(w_j^{(\ell)\top} \mathbf{h}^{(\ell-1)}(x)), \quad \mathbf{h}^{(\ell)}(x') = \frac{1}{\sqrt{n_\ell}} \sum_{j=1}^{n_\ell} A_j^{(\ell)} \boldsymbol{\sigma}(w_j^{(\ell)\top} \mathbf{h}^{(\ell-1)}(x')),$$

where $A_j^{(\ell)} \in \mathbb{R}^r$ is the (random) readout column and $w_j^{(\ell)}$ is the frozen random feature direction. Assume the standard RF-LR initialization: $\{A_j^{(\ell)}\}_{j=1}^{n_\ell}$ are i.i.d. centered with $\mathbb{E}[A_j^{(\ell)} A_j^{(\ell)\top}] = \sigma_A^2 I_r$, independent of $\{w_j^{(\ell)}\}_{j=1}^{n_\ell}$, and take the sequential infinite-width limit $n_\ell \rightarrow \infty$ (Definition 2).

Condition on $\mathbf{h}^{(\ell-1)}(x), \mathbf{h}^{(\ell-1)}(x')$. As $n_\ell \rightarrow \infty$, a conditional law of large numbers yields the usual ReLU Gaussian expectations for the (normalized) inner product and norms, and the isotropic factor $\mathbb{E}\|A_j^{(\ell)}\|^2$ cancels in the cosine ratio; thus the induced correlation recursion is the ReLU cosine map ϱ (for finite r , fluctuations occur at scale $O(r^{-1/2})$; see Appendix C.2).

Step 2: reduce to iterates of ω . With $z_k = (1 - \rho_k)/2$ and $w_k = z_k^{-1/2}$, we have $z_k = \zeta(z_{k-1})$ and therefore $w_k = \omega(w_{k-1})$, i.e. $w_k = \omega^{\circ(k-1)}(w_1)$.

Step 3: asymptotics of $\omega^{\circ k}$. The iterate asymptotic is available in (Terjék and González-Sánchez, 2025, Proposition 3). ■

D.2. RF-LR NTK: recursion-driven depth scaling (not the full-MLP closed form)

For fully trained MLPs at the edge of chaos, the limiting NTK admits an explicit expression in terms of the cosine map and its derivative (Terjék and González-Sánchez, 2025, Proposition 4). This relies on training all weight layers. In RF-LR, only the readouts $A^{(\ell)}$ (and biases $c^{(\ell)}$) are trained and the features $w^{(\ell)}$ are frozen; consequently the RF-LR NTK is described by our recursion (Theorem 4) and its closed form (Corollary 5, Eq. (7)) in terms of $\Sigma^{(\ell)}$ and $\dot{\Sigma}^{(\ell)}$.

The inverse cosine distance framework is useful here at the level of controlling the depth evolution of correlations (hence of $\Sigma^{(\ell)}$ and $\dot{\Sigma}^{(\ell)}$), rather than by importing the fully trained NTK identity.

D.3. Inverse cosine distance matrices and spectral bounds

To formulate dataset-level depth scaling, we work with the deterministic correlation recursion (the mean/infinite-width idealization). Given a dataset $\{x_1, \dots, x_n\} \subset \mathbb{R}^d$, set

$$\rho_{1,ij} = \frac{\langle x_i, x_j \rangle}{\|x_i\| \|x_j\|} \in [-1, 1], \quad \rho_{k,ij} = \varrho(\rho_{k-1,ij}), \quad k \geq 2.$$

Define the layer- k inverse cosine distance matrix $W_k \in \mathbb{S}^n$ by

$$(W_k)_{i,i} = 0, \quad (W_k)_{i_1, i_2} = \left(\frac{1 - \rho_{k, i_1 i_2}}{2} \right)^{-1/2}, \quad i_1 \neq i_2,$$

so that W_k is the inverse cosine distance matrix associated to ρ_k .

Proposition 33 (Spectral bounds for RF-LR inverse cosine distance matrices) *Assume EOC initialization and a dataset with no parallel pairs. For each $k \in [1 : L]$ there exists $\widehat{W}_k \in (1, \infty)$ with $\widehat{W}_k = \Theta_{W_1}(1)$ such that*

$$\|W_k - \omega^{\circ(k-1)}(\widehat{W}_k) (\mathbf{1}_n \mathbf{1}_n^\top - I_n)\| \leq O(\Delta_\phi n^{-1} k) + O(1).$$

Proof This follows from (Terjék and González-Sánchez, 2025, Proposition 6) applied to the deterministic recursion $\rho_{k,ij} = \varrho(\rho_{k-1,ij})$. The argument relies on convexity/monotonicity of ω and standard matrix comparison estimates. ■

D.4. Preliminaries and full statements for Section 4.1

Scalar ReLU kernels as functions of cosine similarity. For unit-variance ReLU features, write $\theta = \arccos(\rho)$. The normalized base and derivative kernels are

$$s(\rho) = \frac{1}{2\pi} ((\pi - \theta) \cos \theta + \sin \theta), \quad \dot{s}(\rho) = \frac{1}{2} - \frac{\theta}{2\pi}.$$

Near $\rho = 1$, parameterize by the inverse cosine distance $w = ((1 - \rho)/2)^{-1/2}$, so that $\rho = 1 - 2w^{-2}$ and $\theta = 2w^{-1} + O(w^{-3})$. Then

$$\dot{s}(\rho) = \frac{1}{2} - \frac{1}{\pi w} + O(w^{-3}), \quad s(\rho) = \frac{1}{2} - \frac{1}{w^2} + O(w^{-3}). \quad (78)$$

Deterministic proxy RF-LR entrywise recursion. Fix an input pair (x, x') with input cosine similarity $\rho_1 \in (-1, 1)$. Define a deterministic correlation recursion $\rho_k = \varrho(\rho_{k-1})$ and $w_k = ((1 - \rho_k)/2)^{-1/2}$. Define the scalar deterministic proxy RF-LR NTK by

$$\Theta^{(1)}(\rho_1) = 1 + s(\rho_1), \quad \Theta^{(k)}(\rho_1) = 1 + \frac{1}{r} \Theta^{(k-1)}(\rho_1) \dot{s}(\rho_k) + \frac{1}{r} s(\rho_k), \quad k \geq 2. \quad (79)$$

This recursion should be viewed as a deterministic proxy: it is obtained from Theorem 4 by replacing the random fields $\Sigma^{(k)}, \dot{\Sigma}^{(k)}$ with their scalar ReLU dual functions $s(\cdot), \dot{s}(\cdot)$ evaluated along the deterministic correlation coefficients ρ_k .

Full statement of Proposition 8. Fix $L \geq 2$ and $r > 1$. Consider a dataset $\{x_i\}_{i=1}^n$ with $\|x_i\| = 1$ and $\rho_{1,ij} = \rho_0 \in (-1, 1)$ for all $i \neq j$. Let

$$K_{ij}^{(L)} := \Theta^{(L)}(\rho_{1,ij}), \quad H := I_n - \frac{1}{n} \mathbf{1}\mathbf{1}^\top, \quad K_c^{(L)} := H K^{(L)} H.$$

Then $K^{(L)} = \Theta^{(L)}(1) I_n + \Theta^{(L)}(\rho_0) (\mathbf{1}\mathbf{1}^\top - I_n)$. In particular, the nonzero eigenvalues of $K_c^{(L)}$ are all equal to

$$\lambda_\perp(K^{(L)}) = \Theta^{(L)}(1) - \Theta^{(L)}(\rho_0),$$

with multiplicity $n - 1$. Moreover, by Theorem 7, there exist constants $0 < c_\Delta < C_\Delta < \infty$ (depending on r, ρ_0) such that for all sufficiently large L ,

$$\frac{c_\Delta}{rL} \leq \lambda_\perp(K^{(L)}) \leq \frac{C_\Delta}{rL}.$$

Full statement of Proposition 9. Under the assumptions above, the eigenvalues of the uncentered Gram matrix $K^{(L)}$ are explicit:

$$\lambda_1(K^{(L)}) = \Theta^{(L)}(1) + (n-1)\Theta^{(L)}(\rho_0), \quad \lambda_2(K^{(L)}) = \dots = \lambda_n(K^{(L)}) = \Theta^{(L)}(1) - \Theta^{(L)}(\rho_0).$$

In particular, $\lambda_1(K^{(L)})$ is a rank-one spike aligned with $\mathbf{1}$, while the spectrum on $\mathbf{1}^\perp$ equals $\lambda_\perp(K^{(L)})$. Moreover, as $L \rightarrow \infty$ (with r fixed) the spike saturates:

$$\lambda_1(K^{(L)}) = n \Theta_\star(r) + O\left(\frac{n}{L}\right), \quad \Theta_\star(r) = \frac{2r+1}{2r-1} = 1 + O\left(\frac{1}{r}\right),$$

and $\lambda_\perp(K^{(L)}) = \Theta(1/(rL))$. Thus, in this model the depth/rank scaling is

$$\lambda_{\max}(K^{(L)}) \asymp n \left(1 + \frac{1}{r}\right), \quad \lambda_{\min}(H K^{(L)} H) \asymp \frac{1}{rL}.$$

D.5. Proof of Theorem 7

Proof

Step 1: correlation alignment. By Proposition 32, there exists a constant $C_w < \infty$ such that for all $k \geq 1$,

$$w_k = \omega^{\circ(k-1)}(w_1) := w_1 + c_0(k-1) + c_1 \log(c_2 w_1 + k-1) + \varepsilon_k, \quad |\varepsilon_k| \leq C_w,$$

where $c_0 = \Delta_\phi \frac{4}{3\pi} > 0$, $c_1 = \Delta_\phi \frac{2}{\pi}$, $c_2 = \Delta_\phi^{-1} \frac{3\pi}{4}$. In particular, $w_k \geq w_1 + c_0(k-1) - C_w$, so choosing k_0 large enough gives $w_k \geq \frac{c_0}{2}k$ for all $k \geq k_0$. Similarly $w_k \leq c_+k$ for some $c_+ < \infty$ and all $k \geq k_0$. Thus $c_-k \leq w_k \leq c_+k$ for $k \geq k_0$, and $1 - \rho_k = 2w_k^{-2} = O(k^{-2})$.

Step 2: kernel saturation. Write $\Theta_k = \Theta^{(k)}(\rho_1)$ and rewrite (79) as

$$\Theta_k = b_k + a_k \Theta_{k-1}, \quad a_k = \frac{1}{r} \dot{s}(\rho_k), \quad b_k = 1 + \frac{1}{r} s(\rho_k).$$

For the limiting constant-coefficient recursion (corresponding to $\rho_k \rightarrow 1$), we have $a = \frac{1}{r} \dot{s}(1) = \frac{1}{2r}$, $b = 1 + \frac{1}{r} s(1) = 1 + \frac{1}{2r}$, hence $\Theta_\star(r) = b/(1-a) = (2r+1)/(2r-1)$.

Set $e_k = \Theta_k - \Theta_\star(r)$. Then

$$e_k = a_k e_{k-1} + (b_k - b) + (a_k - a) \Theta_\star(r).$$

Using (78) and $w_k \geq c_-k$ for $k \geq k_0$, there exists $C < \infty$ such that for all $k \geq k_0$,

$$|a_k - a| := \frac{1}{r} \left| \dot{s}(\rho_k) - \frac{1}{2} \right| \leq \frac{C}{r} \frac{1}{w_k} \leq \frac{C}{r} \frac{1}{k}, \quad |b_k - b| := \frac{1}{r} \left| s(\rho_k) - \frac{1}{2} \right| \leq \frac{C}{r} \frac{1}{w_k^2} \leq \frac{C}{r} \frac{1}{k^2}.$$

Moreover $0 \leq a_k \leq a = 1/(2r) < 1$. Hence for all $k \geq k_0$,

$$|e_k| \leq a |e_{k-1}| + \frac{C'}{k}$$

for some $C' < \infty$. A standard induction (discrete Grönwall) yields $|e_k| \leq C_\Theta/k$ for all $k \geq 1$, in particular $e_k \rightarrow 0$ and $\Theta_k \rightarrow \Theta_\star(r)$.

To make the $O(1/k)$ rate explicit, fix $k \geq k_0$ and iterate the one-step inequality:

$$|e_k| \leq a^{k-k_0} |e_{k_0}| + \sum_{j=k_0+1}^k a^{k-j} \frac{C'}{j}.$$

Since $j \geq k_0 + 1$ implies $1/j \leq 1/k_0$ and also $j \leq k$ implies $1/j \leq 1/k$, we can bound

$$\sum_{j=k_0+1}^k a^{k-j} \frac{1}{j} \leq \frac{1}{k} \sum_{t=0}^{k-k_0-1} a^t \leq \frac{1}{k} \cdot \frac{1}{1-a}.$$

Also $a^{k-k_0} |e_{k_0}| \leq |e_{k_0}| \leq (k_0 |e_{k_0}|)/k$. Therefore for all $k \geq k_0$,

$$|e_k| \leq \frac{1}{k} \left(k_0 |e_{k_0}| + \frac{C'}{1-a} \right).$$

Absorbing the finitely many values $k < k_0$ into the constant gives $|e_k| \leq C_\Theta/k$ for all $k \geq 1$, as claimed.

Step 3: diagonal–off-diagonal gap. Let $\Theta_k^{\text{diag}} = \Theta^{(k)}(1)$ and $\Theta_k^{\text{off}} = \Theta^{(k)}(\rho_1)$, and set $\Delta_k = \Theta_k^{\text{diag}} - \Theta_k^{\text{off}} \geq 0$. The diagonal recursion is $\Theta_k^{\text{diag}} = b + a \Theta_{k-1}^{\text{diag}}$, while the off-diagonal recursion is $\Theta_k^{\text{off}} = b_k + a_k \Theta_{k-1}^{\text{off}}$. Subtracting gives the exact identity

$$\Delta_k := a \Delta_{k-1} + (a - a_k) \Theta_{k-1}^{\text{off}} + (b - b_k).$$

For k large, $\Theta_{k-1}^{\text{off}}$ is bounded above and below by positive constants (since $\Theta_k^{\text{off}} \rightarrow \Theta_\star(r) \in (1, \infty)$). Also by (78) and $w_k \asymp k$,

$$a - a_k = \frac{1}{r} \left(\frac{1}{2} - \dot{s}(\rho_k) \right) = \Theta \left(\frac{1}{r w_k} \right) = \Theta \left(\frac{1}{r k} \right), \quad b - b_k = \frac{1}{r} \left(\frac{1}{2} - s(\rho_k) \right) = O \left(\frac{1}{r w_k^2} \right) = O \left(\frac{1}{r k^2} \right).$$

Thus there exist k_1 and constants $0 < c < C < \infty$ such that for all $k \geq k_1$,

$$a \Delta_{k-1} + \frac{c}{r k} \leq \Delta_k \leq a \Delta_{k-1} + \frac{C}{r k}.$$

Iterating these comparison recursions yields $c_\Delta/(rk) \leq \Delta_k \leq C_\Delta/(rk)$ for all $k \geq k_1$, completing the proof. $\blacksquare$

D.6. Proof of Proposition 8

Proof The explicit form of $K^{(L)}$ follows from $\rho_{1,ii} = 1$ and $\rho_{1,ij} = \rho_0$ for $i \neq j$. The matrix $\mathbf{1}\mathbf{1}^\top - I_n$ has eigenvalue $n - 1$ along $\mathbf{1}$ and eigenvalue -1 on $\mathbf{1}^\perp$. Therefore $K^{(L)}$ has eigenvalue $\Theta^{(L)}(1) + (n - 1)\Theta^{(L)}(\rho_0)$ along $\mathbf{1}$ and eigenvalue $\Theta^{(L)}(1) - \Theta^{(L)}(\rho_0)$ on $\mathbf{1}^\perp$. Since H is the orthogonal projection onto $\mathbf{1}^\perp$, the nonzero eigenvalues of $K_c^{(L)} = H K^{(L)} H$ are exactly the $\mathbf{1}^\perp$ eigenvalues of $K^{(L)}$, with multiplicity $n - 1$. The $1/(rL)$ bounds then follow from Theorem 7 applied to $\rho_1 = \rho_0$. $\blacksquare$

D.7. Proof of Proposition 9

Proof The eigenvalue identities are the $\mathbf{1}$ and $\mathbf{1}^\perp$ eigenvalues established in the proof of Proposition 8. The constant-mode saturation follows from Theorem 7: both $\Theta^{(L)}(1)$ and $\Theta^{(L)}(\rho_0)$ converge to $\Theta_\star(r)$ at rate $O(1/L)$, hence

$$|\lambda_1(K^{(L)}) - n\Theta_\star(r)| \leq |\Theta^{(L)}(1) - \Theta_\star(r)| + (n - 1)|\Theta^{(L)}(\rho_0) - \Theta_\star(r)| := O\left(\frac{n}{L}\right).$$

Finally, $\lambda_{\max}(K^{(L)}) \asymp n(1 + 1/r)$ follows from $\lambda_1(K^{(L)}) \sim n\Theta_\star(r)$ and $\Theta_\star(r) = (2r + 1)/(2r - 1) = 1 + O(1/r)$, while $\lambda_{\min}(H K^{(L)} H) = \lambda_\perp(K^{(L)}) \asymp 1/(rL)$ follows from Proposition 8. $\blacksquare$

D.8. Remark: proxy lower bounds vs positivity (smallest eigenvalue)

Remark 34 (Can the smallest eigenvalue become negative?) *No: for any finite network, the empirical NTK Gram matrix $\hat{K}$ at initialization is a Gram matrix of gradients, $\hat{K}_{ij} = \langle \nabla_\theta f(x_i), \nabla_\theta f(x_j) \rangle$, hence $\hat{K} \succeq 0$ deterministically and all its eigenvalues are ≥ 0 . Moreover, centering preserves positive semidefiniteness: $H \hat{K} H \succeq 0$ since H is an orthogonal projection.*

What can happen is that the deterministic proxy lower bound (e.g. $\lambda_{\min}(HK^{(L)}H) \asymp 1/(rL)$ in Proposition 8) is smaller than the magnitude of random fluctuations. In general, Weyl’s inequality gives

$$\lambda_{\min}(H\hat{K}H) \geq \lambda_{\min}(HKH) - \|H(\hat{K} - K)H\|,$$

so if $\|H(\hat{K} - K)H\|$ is larger than $1/(rL)$, this bound becomes vacuous (it may be negative), but the true eigenvalue still satisfies $\lambda_{\min}(H\hat{K}H) \geq 0$. Thus, the relevant concern is not “going below zero” but rather “being driven close to zero” when the $1/(rL)$ signal is swamped by noise.

D.9. Remark: infinite-depth limit after centering

Remark 35 (Infinite-depth limit: existence and degeneracy after centering) *The correlation recursion satisfies $1 - \rho_k = O(k^{-2})$ as in full-width MLPs at the EOC Terjék and González-Sánchez (2025); Hayou et al. (2022), so deep features become increasingly aligned. For the mean RF-LR NTK recursion (79), Theorem 7 implies that for any fixed $\rho_1 \in (-1, 1)$,*

$$\Theta^{(L)}(\rho_1) \rightarrow \Theta_{\star}(r) \quad \text{and} \quad \Theta^{(L)}(1) \rightarrow \Theta_{\star}(r),$$

so the entrywise infinite-depth limit of the (mean) Gram matrix is a constant kernel: $K^{(L)} \rightarrow \Theta_{\star}(r) \mathbf{1}\mathbf{1}^{\top}$. Consequently, after centering one has $HK^{(L)}H \rightarrow 0$, and the relevant spectrum on $\mathbf{1}^{\perp}$ collapses. In particular, in the equicorrelated model Proposition 8 gives $\lambda_{\min}(HK^{(L)}H) = \Theta(1/(rL))$, so a non-degenerate deep limit on $\mathbf{1}^{\perp}$ can only hold under a joint scaling (e.g. r growing with L) or after an explicit rescaling, such as considering $rL HK^{(L)}H$, whose nonzero eigenvalues are $rL \lambda_{\perp}(K^{(L)}) \in [c_{\Delta}, C_{\Delta}]$ for all large L .